Geometry-First Mathematics

Image Prompt Blueprint v6

Constraint-stack production prompts for GPT Image 1.5

265 prompt cards | 20-prompt pilot | 384 recommended full-library variants

Operating rule	Prompt lines are behavior constraints, not mood prose.
Prompt order	Scene -> Subject -> Conceptual action -> Composition -> Materials / medium -> Surface / text policy -> Use
Math precision	Generated art creates atmosphere and conceptual apparatus. Exact math is vector overlay.
Pilot first	Run the 20-prompt pilot before spending on the full library.

1. What changed from v5

• Prompt lines are constraints, not mood prose. Each line controls scene, subject, action, composition, materials, surface policy, or intended use.

• The prompt order is fixed: Scene -> Subject -> Conceptual action -> Composition -> Materials / medium -> Surface / text policy -> Use.

• Use concrete nouns and exact counts; avoid vague quantities like many, several, countless, or a few.

• One conceptual move per instructional image. Opening spreads may use station-based friezes with equal cells and blank gutters.

• Generated art carries atmosphere, apparatus, and conceptual gesture. Exact labels, axes, equations, theorem lines, and tick marks are vector overlays.

• Historical plates use instruments first. Closed books or blank pages may appear; open manuscripts with fake script are avoided by construction.

• Recurring motifs are generated as reference assets first and reused through image edits to prevent style drift.

2. API defaults and production parameters

Model	gpt-image-1.5
Prompt cards	265
Pilot cards	20
Full variants	384
Output format	PNG
Sizes used	1024x1024, 1536x1024
Qualities used	high, medium

3. Reference-image workflow

Step 1 Generate System Assets: SYS.1 brass_unit_stick, SYS.2 amber_invariant_glow, SYS.3 refinement_lens

Step 2 Lock Asset Selection: Choose one approved asset for each motif. Do not regenerate recurring props independently in later chapters.

Step 3 Use Image Edit For Callbacks: When a prompt has reference_tags, use the approved system asset as an input image and write edit prompts that preserve material, proportions, patina, and silhouette.

Edit Invariant Sentence: Change only the surrounding scene and conceptual apparatus. Preserve the referenced motif shape, material, scale character, patina, and lighting logic.

4. Batch QA protocol

• Run the 20-prompt pilot first with three variants each using gpt-image-1.5, quality high for pilot regardless of record default.

• Reject any variant with generated letters, numerals, formulas, pseudo-script, signage, visible watermarks, or fake diagram labels.

• Reject any variant where the focal hierarchy is ambiguous after a two-second look.

• Reject any instructional plate with more than one conceptual action unless it is explicitly a two-panel, three-panel, or frieze prompt.

• Run automated OCR/text detection where possible; treat OCR hits as review flags, not final judgment.

- Review side-by-side by section. Pick one winning variant or rewrite the prompt before moving to the next section cluster.
- Only after the pilot passes, batch-generate the rest in section order.

5. Pilot set

Generate these first with three variants each at high quality. Revise the global rules before full production if the pilot exposes a systemic failure.

Prompt ID	Section	Title	Family
GFV6-FM_1-A	FM.1	What this revision thinks the original manuscript is really trying to do	Modern map and recovery
GFV6-FM_5-A	FM.5	The permanent scaffold	Modern map and recovery
GFV6-0_5-A	0.5	What school usually hides here	Tactile workshop realism
GFV6-0_7-A	0.7	Thought experiments	Tactile workshop realism
GFV6-1_2-A	1.2	Directed segments: where negatives actually come from	Tactile workshop realism
GFV6-1_5-A	1.5	Historical note: why negatives felt suspicious	Tactile workshop realism
GFV6-4_2-A	4.2	The Eudoxus definition	Vellum-and-instrument geometry
GFV6-4_3-A	4.3	A concrete pass through the Eudoxus test	Vellum-and-instrument geometry
GFV6-5_3-A	5.3	The similar-triangle multiplication machine	Vellum-and-instrument geometry
GFV6-6-A	6	Division as inverse scaling	Vellum-and-instrument geometry
GFV6-6_1-B	6.1-B	Division roles - restrained triptych	Vellum-and-instrument geometry
GFV6-6_5-A	6.5	Slope is the geometric picture of rate	Vellum-and-instrument geometry
GFV6-10_1-A	10.1	The tower of operations	Scale-gate and odometer
GFV6-10_3-A	10.3	The exponent laws are structural facts	Scale-gate and odometer
GFV6-10_4-A	10.4	Fractional exponents are forced, not invented for mischief	Scale-gate and odometer
GFV6-10_5-A	10.5	Logarithms count multiplicative steps	Scale-gate and odometer
GFV6-11_5-A	11.5	Connection back to Eudoxus	Refinement optics
GFV6-12_7-A	12.7	Euler's formula as compressed geometry	Transformation theatre
GFV6-16_2-A	16.2	Derivative: the slope that survives zooming in	Refinement optics
GFV6-16_8-A	16.8	The Fundamental Theorem of Calculus	Refinement optics

6. Visual families

Tactile workshop realism

Purpose	Magnitudes, units, measurement, number line, early fractions. Rods, string, pins, vellum, calipers, hands, shadows.
Scene default	quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Materials default	editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
API default	quality medium; size 1536x1024

Vellum-and-instrument geometry

Purpose	Eudoxus, intercept theorem, similarity, ratio machines. Drafting arms, balances, taut threads, transparent triangles.
Scene default	drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled studio light
Materials default	photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension, glass reflections
API default	quality high; size 1536x1024

Transformation theatre

Purpose	Negatives, complex numbers, functions, matrices, polynomials. Machines, stages, gears, turntables, input/output paths.
Scene default	small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and controlled spotlights
Materials default	editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft smoke-free lighting
API default	quality medium; size 1536x1024

Scale-gate and odometer

Purpose	Exponents and logarithms. Repeated resize portals, ratio shells, counters that count multiplicative steps.
Scene default	clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate crossings
Materials default	premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer wheels, soft studio shadows
API default	quality high; size 1536x1024

Refinement optics

Purpose	Irrationals, limits, calculus, Taylor series. Lenses, nested shutters, microscopes, slices, converging masks.
Scene default	optical workbench with nested shutters, lenses, microscopes,

	translucent windows, and increasingly fine slices
Materials default	macro editorial photography mixed with precise illustration; glass, black velvet, warm amber focal point, clean shadows
API default	quality high; size 1536x1024

Turning and projection observatory

Purpose	Trigonometry and Euler's formula. Rotating arms, shadows, circular rails, sky instruments, periodic motion.
Scene default	observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and sky instruments
Materials default	night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean projected shadows
API default	quality high; size 1536x1024

Field and flow

Purpose	Differential equations, waves, phase portraits. Local arrows, currents, vibrating media, paths obeying local rules.
Scene default	field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-space traces
Materials default	editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
API default	quality high; size 1536x1024

Invariant architecture

Purpose	Relativity, Noether, modern structure. Quiet crystals, observer slices, preserved amber cores, symmetry chambers.
Scene default	quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass measuring structures
Materials default	restrained architectural editorial render; translucent glass, graphite stone, amber thread, sober studio light
API default	quality high; size 1536x1024

Modern map and recovery

Purpose	Front matter, where-to-go-next, appendices. Maps, cabinets, cathedral scaffolds, disciplined visual systems.
Scene default	calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Materials default	museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
API default	quality high; size 1536x1024

7. Prompt cards

Modern map and recovery

GFV6-FM_1-A - FM.1: What this revision thinks the original manuscript is really trying to do

Placement	full-width opening plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: horizontal frieze on a modular studio wall; exactly seven equal stations from left to right
Subject: one amber thread passes through: rod comparison, arrow translation, scale gate, nested shutter, curve
under lens, local field arrows, architectural prism
Conceptual action: show the book spine as one repeated move: object -> action -> relation -> refinement ->
invariant
Composition: seven stations, equal size, separated by blank gutters; each station reads as a simple icon before
it reads as a scene
Materials / medium: graphite wall, brass instruments, vellum, glass lens, matte stone prism, warm amber thread
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The book's thesis is one spine, not a tour of disconnected topics.

Overlay notes: Overlay optional: small chapter waypoint labels on the ascending path. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_1-B - FM.1-B: Thesis spine - seven-panel reference frieze

Placement	companion reference frieze
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: wide modular frieze with exactly seven square cells
Subject: one brass bead travels cell by cell through comparison, translation, scaling, refinement, calculus zoom,
field rule, invariant prism
Conceptual action: same object survives the whole curriculum by changing the operation around it
Composition: seven cells, equal size, blank gutters, bead aligned on one horizontal path
Materials / medium: wood, brass, vellum, glass lens, water field, graphite stone prism, amber thread
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The book is a connected spine: compare, move, resize, refine, approximate, flow, preserve.

Overlay notes: Use as a foldout or section divider. Add captions after generation only. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_2-A - FM.2: How to use this book if school made mathematics feel fake

Placement	margin or opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a quiet desk scene where a stack of rote worksheets has been pushed aside and replaced by a compass,
string, brass unit rod, transparent tracing paper, and a small lamp illuminating one simple segment.
Conceptual action: The reader is being respected as a rebuilding learner; the subject is being rebuilt honestly;
the mood feels like leaving a fluorescent classroom and entering a real workshop
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
```

```
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The reader is not being remediated; the subject is being rebuilt honestly.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_3-A - FM.3: A decoding tool for every new idea

Placement	one-page reference image
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3

```
Scene: single brass bead on a clean tabletop passing through exactly four square lenses
Subject: the same bead appears as picture, action, relation, and compression while identity is preserved
Conceptual action: each lens changes the learner’s view, while object identity is preserved
Composition: four lenses in a cross around the bead; center object remains aligned; empty corners for later
labels
Materials / medium: clear glass lenses, brass bead, vellum shadows, warm tabletop side light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Picture, action, relation, compression are four views of the same idea.

Overlay notes: Overlay recommended: the four words can be added in layout. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_4-A - FM.4: A visual grammar for the rest of the book

Placement	spread opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: an elegant wall of six framed visual motifs: layered stack, path, resize, turn, constraint, and
refinement.
Conceptual action: The reader learns to recognize the book's recurring visual families; each frame is simple and
iconic but rendered as fine art objects: translucent layers, a ghost path, a dilation shadow, a rotating beam, a
glowing selected locus, and nested shutters closing
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The reader learns to recognize the book's recurring visual families.

Overlay notes: Overlay: add motif names as typography after generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_5-A - FM.5: The permanent scaffold

Placement	full-width anchor plate
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3

```
Scene: long workbench divided into exactly six equal stations
Subject: one plain object is moved, resized, turned, partitioned, refined, and tested for what stays fixed
Conceptual action: make the permanent scaffold visible as repeated actions on the same object
Composition: six stations in a straight row; one action per station; station boundaries are blank gutters, gutter
boundaries
Materials / medium: wood bench, brass unit stick, small turntable, scale gate, nested glass shutter, amber
invariant thread
```

```
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Every later concept is a reuse of the same acts.

Overlay notes: Overlay recommended for platform names and chapter references. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_5-B - FM.5-B: Permanent scaffold - simplified learner card

Placement	companion learner card
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: single square learning card on a clean desk
Subject: four empty compartments arranged as a 2 by 2 grid: object, operation, equivalence, invariant
Conceptual action: show the scaffold as four reusable roles in a compact learning card
Composition: exactly four compartments; one plain object appears in all four; large blank heading space
Materials / medium: matte card, brass bead, vellum overlay, amber thread, soft desk light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: A simpler scaffold card for readers who need the map before the machinery.

Overlay notes: Add four word labels as vector type only. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-FM_6-A - FM.6: A roadmap of the journey

Placement	front matter spread
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a long road that begins with two unmeasured sticks and passes through milestones: unit choice, number
line, partition, ratio loom, dilation tower, turning observatory, function city, calculus microscope, wave
landscape, spacetime arch, and modern mathematics horizon.
Conceptual action: The book is a spiral ascent, a spiral ascent; each milestone is recognizable but clear and
readable
Composition: wide left-to-right route with exactly seven milestones; each milestone is visually simple; open
sky/space above for captions
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The book is a spiral ascent, not a checklist.

Overlay notes: Overlay recommended for part names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Tactile workshop realism

GFV6-0-A - 0: Geometry before numbers: magnitudes and sameness

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: two handmade wooden rods on a table under soft side light.
Conceptual action: Magnitude and sameness come before measurement; one rod aligns exactly with a transparent
ghost of the other after a slide-and-rotate motion; In the background, a measuring scale remains blurred and
secondary
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Magnitude and sameness come before measurement.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_1-A - 0.1: Magnitudes are older than numerals

Placement	subsection side image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a rope pulled taut between two pins, a carved stick, and the edge of a stone slab.
Conceptual action: Length exists before its numeric report; endpoints are subtly highlighted; a faint translucent
line connects the examples as instances of extension
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Length exists before its numeric report.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_2-A - 0.2: Congruence: sameness without counting

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: two segments on separate sheets of translucent vellum.
Conceptual action: Sameness is what survives rigid motion; one sheet has been slid, rotated, and flipped in three
ghosted positions until the segments land in perfect overlap
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Sameness is what survives rigid motion.

Overlay notes: Overlay optional: endpoint names and motion arrows. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_3-A - 0.3: The first honest act of measurement: choose a unit

Placement	conceptual pipeline

Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: four calm stages from left to right: raw segment, chosen brass unit stick, repeated comparison of the
unit against the segment, and a small blank tag waiting for the measured report.
Conceptual action: Measurement is a relationship between a magnitude and a chosen reference; the tag is visually
last and least physical
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Measurement is a relationship between a magnitude and a chosen reference.

Overlay notes: Overlay recommended for stage labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_4-A - 0.4: Comparison before multiplication

Placement	small horizontal plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: three rods: A and B are placed end-to-end to form a combined length; C sits below as the target
comparison.
Conceptual action: Comparison and concatenation are enough for the first layer of mathematics; use shadows and
alignment to make longer, shorter, and equal feel decidable using alignment rather than generated numbers
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Comparison and concatenation are enough for the first layer of mathematics.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_5-A - 0.5: What school usually hides here

Placement	split-panel corrective image
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a calm corrective split image.
Conceptual action: Most early confusion comes from collapsing four layers into one; on the left, show one
confusing heap where point, segment, unit stick, arrow, and report token have been collapsed together
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Most early confusion comes from collapsing several layers into one.

Overlay notes: Overlay can add layer names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_6-A - 0.6: Historical note: why the Greeks cared so much

Placement	vertical historical tableau
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: survey ropes, limestone blocks, a wax tablet, and a square whose diagonal refuses to match a neat row of
counting pebbles.
Conceptual action: Incommensurable lengths forced mathematics toward proof; two hands hover over the diagonal,
with the seriousness of people realizing computation alone fails
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Incommensurable lengths forced mathematics toward proof.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_7-A - 0.7: Thought experiments

Placement	margin prompt image
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: the same unchanged object measured twice with two different unit sticks: one short brass unit repeated
eight times, one longer wood unit repeated four times.
Conceptual action: Run the idea with your hands before accepting the notation; the object remains physically
identical in the center while the reports are represented only by two blank tags waiting for overlay
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Run the idea with your hands before accepting the notation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-0_8-A - 0.8: Common confusion to kill early

Placement	corrective plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a table edge seen through three translucent layers.
Conceptual action: A thing, a standard, and a number are related but distinct; bottom layer: the actual edge
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
```

```
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: A thing, a standard, and a number are related but not identical.

Overlay notes: Overlay optional: layer names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1-A - 1: The number line as geometry: oriented length

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a bare road becoming a coordinate line only after three deliberate choices: a stake hammered as origin,
an arrow of travel painted on the ground, and a unit stride marked by a brass rod.
Conceptual action: A number line is a chosen frame placed on a line; the world looks unchanged before and after;
only the reference frame has been installed
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: A number line is a chosen frame placed on a line.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_1-A - 1.1: A line becomes a number line only after choices

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a plain line on a drafting table as a stage before performance.
Conceptual action: Origin, direction, and unit turn a line into a number line; a hand places a small pin for
origin, a compass arrow sets positive direction, and a unit tile is laid down
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Origin, direction, and unit turn a line into a number line.

Overlay notes: Overlay recommended for O, direction arrow, and unit. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_2-A - 1.2: Directed segments: where negatives actually come from

Placement	horizontal plate
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a fixed origin bead on a horizontal rail.
Conceptual action: Negative means opposite direction, opposite orientation; one arrow of a certain length points right; a matching ghost arrow of exactly the same length points left
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Negative means opposite direction, not less-than-real.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_3-A - 1.3: Integers as repeated unit translations

Placement	small sequence image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a brass unit block being copied along a rail like identical train cars.
Conceptual action: Integers record repeated unit moves; each copy is a completed translation from the last endpoint
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Integers record repeated unit moves.

Overlay notes: Overlay optional: integer coordinates after generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_4-A - 1.4: Another lens: balance sheets and elevators

Placement	triptych
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a restrained triptych of orientation analogies with Each panel uses the same visual grammar: a reference, a direction, and equal distance in opposite orientations.
Conceptual action: Signed quantities are reference-relative orientations across domains.
Composition: exactly three equal panels with blank gutters; one conceptual role per panel; each panel contains at most two main objects
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Signed quantities are reference-relative orientations across domains.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_5-A - 1.5: Historical note: why negatives felt suspicious

Placement	historical note plate
Priority	core / regeneration high

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: a historical merchant desk with physical tokens: ordinary assets sit above a reference line, debts hang
below it like counterweights.
Conceptual action: Negatives were suspect until number could mean directed relation; one older ledger treats the
below-line tokens as suspicious shadows; a newer geometric overlay shows them simply as opposite-directed
quantities
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Negatives were suspect until number could mean directed relation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_6-A - 1.6: The dimension subtlety

Placement	layered explanatory plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: one fixed point on a blank line, its directed segment from a chosen origin, and a separate blank
coordinate tag hovering above like a report.
Conceptual action: A coordinate is a report; a directed segment is a geometric object; the physical arrow and the
abstract tag different materials
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: A coordinate is a report; a directed segment is a geometric object.

Overlay notes: Overlay recommended for dimension notes. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_7-A - 1.7: Thought experiments

Placement	margin visual
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a path with footsteps measured by two different reference sticks placed beside it.
Conceptual action: Change the frame and the report changes; the physical place stays fixed; one reference stick
is short and repeats eight times; the other is longer and repeats four times
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Change the frame and the report changes; the place does not.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-1_8-A - 1.8: A preview of later structure

Placement	motif bridge
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a single luminous line visited by three hovering transformation glyphs: slide, reflect, scale.
Conceptual action: The number line is already a symmetry space; each glyph leaves a different trace while a small invariant glows beneath it: difference, distance, ratio
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: The number line is already a symmetry space.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2-A - 2: Addition and subtraction as geometry of motion

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a figureless sequence of arrows moving across a clean floor: first arrow, second arrow, final displacement.
Conceptual action: Addition is the geometry of chaining moves; the final arrow floats above the two-step path like the compressed summary of the journey
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative space for title overlay
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Addition is the geometry of chaining moves.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_1-A - 2.1: Addition is composition of translations

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a point traveling from P to Q to R as two physical arrow moves, with a single larger translucent arrow spanning start to finish.
Conceptual action: The sum is the one move equivalent to two moves in sequence; the intermediate point is visible but visually subordinate to the total displacement
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
```

```
Use: pedagogical editorial plate for an adult math book
```

Caption: The sum is the one move equivalent to two moves in sequence.

Overlay notes: Overlay for point names and arrow labels recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_2-A - 2.2: Why commutativity is not a coincidence

Placement	hybrid diagram
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: two arrows as colored glass strips forming adjacent sides of a parallelogram.
Conceptual action: Commutativity is a symmetry of translation in flat space; a diagonal beam reaches the same far
corner by either route; ghosted footsteps trace both orders
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Commutativity is a symmetry of translation in flat space.

Overlay notes: Overlay optional: a+b and b+a labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_3-A - 2.3: Subtraction is composition with the inverse move

Placement	small motion plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a bright arrow move followed by a translucent reversed arrow that perfectly unwinds it, like a path
being rewound.
Conceptual action: Subtracting means adding the undoing move; the endpoint returns to the origin marker
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Subtracting means adding the undoing move.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_4-A - 2.4: Associativity: grouping does not change the total path

Placement	minimal route map
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a three-segment route across a map with two faint transparent grouping bubbles placed differently over
the same route.
```

```
Conceptual action: Grouping changes description, description while accumulated displacement stays fixed; the physical path and final endpoint remain identical
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Grouping changes description, not accumulated displacement.

Overlay notes: Overlay optional: parentheses as layout type only. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_5-A - 2.5: Another lens: addition as bookkeeping of change

Placement	alternate lens plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a ledger of changes as stacked translucent arrows rather than columns of digits.
Conceptual action: Addition tracks net change across any compatible quantity; gains, losses, deposits, and withdrawals become displacements on a common line
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Addition tracks net change across any compatible quantity.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_6-A - 2.6: Historical note: from counting to structure

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: an abacus, tally marks, and a modern vector diagram connected by a single thread.
Conceptual action: Old arithmetic becomes structural when viewed as transformation; the abacus counts results; the vector diagram reveals structure
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible through instruments
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Old arithmetic becomes structural when viewed as transformation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-2_7-A - 2.7: Thought experiments

Placement	interactive mental scene
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a sparse grid floor with a token moved north then east, and a ghost route east then north to the same square.
Conceptual action: Try the move in both orders and watch the endpoint; include a second tiny scene where a move and its undoing cancel
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space for margin placement
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Try the move in both orders and watch the endpoint.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3-A - 3: Fractions and rationals: partitioning and repeatability

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a long ribbon being divided by parallel light beams into equal parts, with three parts recombined elsewhere into a new segment.
Conceptual action: Fractions are constructed by equal partition and finite repetition; the materials feels tactile: paper, thread, pins, and shadows
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative space for title overlay
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Fractions are constructed by equal partition and finite repetition.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_1-A - 3.1: Equal partition is the key act

Placement	subsection image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a single strip of paper being prepared for equal division, with a fan of auxiliary lines from one endpoint and parallel cuts projected back onto the strip.
Conceptual action: A fraction begins as a construction of equal parts; the focus is the act of forcing equality through geometry, the construction action before symbols
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: A fraction begins as a construction of equal parts.

Overlay notes: Overlay recommended for construction lines. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_2-A - 3.2: One over n is an actual constructed length

Placement	close-up plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: one highlighted piece cut from a unit ribbon after division into equal parts.
Conceptual action: One over n is one constructed equal part of the unit; the other pieces remain pale; the chosen
piece glows as a real little length, a tangible little length
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: One over n is one constructed equal part of the unit.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_3-A - 3.3: m/n is repeated taking of a constructed part

Placement	horizontal sequence
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: identical small unit-fraction tiles being placed end-to-end to build a longer segment.
Conceptual action: m/n is m copies of one equal part; the repeated pieces is visibly congruent, with a hand or
ghost trail indicating finite repetition
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: m/n is m copies of one equal part.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_4-A - 3.4: Equivalent fractions are the same magnitude wearing different descriptions

Placement	split-and-overlay image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: two different tile patterns - one with six small tiles, one with three larger tiles - sliding into
perfect overlap over the same length.
Conceptual action: Equivalent fractions are different reports for one congruent length; the viewer sees different
partitions, same final magnitude
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Equivalent fractions are different reports for one congruent length.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_5-A - 3.5: Commensurable versus incommensurable

Placement	conceptual contrast
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: two rods happily measured by the same small golden bead, then a square diagonal beside them where the
bead never quite fits
Conceptual action: One pair of magnitudes shares a common measure; one diagonal refuses.
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Some magnitudes share a common measure; some refuse.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_6-A - 3.6: Another lens: music and rhythm

Placement	alternate lens plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: a circular rhythm wheel split into equal pulses, with different rhythmic patterns lighting up different
subsets while the cycle stays fixed.
Conceptual action: Fractions are also equal divisions of time and rhythm; let the fraction idea become temporal:
equal beats, repeated beats, same cycle
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: Fractions are also equal divisions of time and rhythm.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_7-A - 3.7: Historical note: why fractions mattered

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: a granary, a field boundary rope, and an astronomical measuring arc sharing the same quiet problem:
finite quantities must be divided and compared.
Conceptual action: Fractions were born from dividing real magnitudes fairly and reliably; the scene connects
land, food, trade, and sky through one practical arrangement rather than a busy history mural
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
```

```
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Fractions were born from dividing real magnitudes fairly and reliably.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_8-A - 3.8: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: one ribbon folded in halves, thirds, and fifths as ghosted alternate constructions, with the same
original ribbon visible underneath.
Conceptual action: Fold first; calculate later; the reader feels invited to imagine folding, cutting, and
recombining rather than computing
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Fold first; calculate later.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-3_9-A - 3.9: Common confusion

Placement	corrective plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: quiet tabletop workshop; wood surface, vellum sheets, pins, string, calipers, and side light
Subject: an elegant image of a unit segment, a part of that segment, and a ratio tag as three separate layers.
Conceptual action: A fraction can name a length, a ratio, or an operator; keep the roles visually separate; in
the corner, a childish pie-chart image is faded out like an insufficient metaphor
Composition: one physical action; top-down or shallow angle; three primary objects maximum; empty margin for
later labels
Materials / medium: editorial still life; brass, wood grain, linen string, matte paper, transparent vellum; 50mm
lens feel; soft contact shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical editorial plate for an adult math book
```

Caption: A fraction can name a length, a ratio, or an operator; do not blur them.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Vellum-and-instrument geometry

GFV6-4-A - 4: Proportion theory: equality of ratios before multiplication

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: two pairs of magnitudes fed into a large elegant comparison loom.
Conceptual action: Proportion is equality of comparison behavior before multiplication exists; whole-number
copies of each magnitude rise and fall on paired balances, and the loom emits matching woven fingerprints when
every comparison agrees
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Proportion is equality of comparison behavior before multiplication exists.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_1-A - 4.1: Why this section is load-bearing

Placement	chapter anchor
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a load-bearing stone bridge where addition/comparison stones form the approach, a central proportion
arch carries the span, and similar-triangle construction rests beyond it.
Conceptual action: Ratio theory carries the weight of later multiplication; leave a transparent gap where the
proportion arch would be; the bridge proportion arch is missing and the bridge fails at that span
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Ratio theory carries the weight of later multiplication.

Overlay notes: Overlay optional for visual layer names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_2-A - 4.2: The Eudoxus definition

Placement	hero diagram-art hybrid
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: drafting table with exactly two small balance scales side by side
Subject: each balance receives paired stacks of segment copies; both balances tip the same way across the same
trial
Conceptual action: teach Eudoxus as identical response to every finite comparison test
Composition: two balances centered; two faint ghost trials behind them; blank top margin for vector overlay
Materials / medium: brass balances, wood segment copies, vellum trial cards with abstract texture only
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Equal ratios answer every whole-number comparison the same way.

Overlay notes: Precise symbols should be overlaid later, not generated. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_2-C - 4.2-C: Eudoxus definition - clean balance companion

Placement	diagram-art hybrid
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: minimal vellum plate for later diagram overlay
Subject: two blank balance scales, two rows of segment-copy stacks, and one amber “same test response” line
Conceptual action: create atmosphere only; exact m, n, A, B, C, D labels are vector overlays
Composition: centered apparatus; large empty margins; apparatus only
Materials / medium: pale vellum, brass balance arms, wooden segment blocks, soft grey shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Same ratio means same response to every finite comparison test.

Overlay notes: Overlay may add one worked m-versus-n test after generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_2-B-A - 4.2-B: The Eudoxus definition - alternate

Placement	optional secondary plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a field of tiny comparison switches as an abstract fingerprint.
Conceptual action: A ratio can be known by its full comparison fingerprint; two different-looking ratio objects
cast the same pattern of lit and unlit switches across the field; exact construction lines are reserved for
vector overlay.
Composition: minimal apparatus centered on blank vellum; 70 percent clean overlay area; one construction scaffold
only
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: A ratio can be known by its full comparison fingerprint.

Overlay notes: Optional overlay: mA vs nB and mC vs nD. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_3-A - 4.3: A concrete pass through the Eudoxus test

Placement	instructional plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: ratio-testing bench with exactly one trial in progress and two faint completed trials behind it
Subject: the same pair of copy counts is applied to two different ratio pairs
Conceptual action: make the learner see “same ratio” as same comparison behavior, same comparison behavior before
division syntax
Composition: foreground trial sharp; background trials translucent; single amber result marker
Materials / medium: wood segment stacks, brass calipers, vellum cards with blank surfaces
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: A single Eudoxus test is just repeated addition plus comparison.

Overlay notes: Overlay for sample m and n recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_4-A - 4.4: Why this is brilliant rather than cumbersome

Placement	conceptual portrait
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a ratio as a key whose teeth are outcomes of every possible comparison.
Conceptual action: The cumbersome test becomes a precise identity fingerprint; two different-looking magnitude
pairs produce identical keys that open the same lock
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: The cumbersome test becomes a precise identity fingerprint.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_5-A - 4.5: Easy case and hard case

Placement	split contrast
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: left side: two lengths measured neatly by a common bead-string.
Conceptual action: Eudoxus works for commensurable and incommensurable cases; right side: a square diagonal
approached by finer bead-strings that never settle into exact fit
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Eudoxus works whether or not a common measure exists.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_6-A - 4.6: The intercept theorem is now earned, not assumed

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: sunlight-like parallel rays cutting across two angled reeds or strings.
Conceptual action: Parallel structure transfers ratios across lines; equal marks on one support project into
proportional marks on the other through clean parallel lines
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
```

```
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Parallel structure transfers ratios across lines.

Overlay notes: Overlay required for exact theorem lines and segment labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_7-A - 4.7: Similarity as same shape, different scale

Placement	visual proof plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: three versions of the same triangular frame at different sizes, nested like architectural models.
Conceptual action: Similarity is preserved shape under scale change; their angles align by transparent overlays
while side lengths grow
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Similarity is preserved shape under scale change.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_8-A - 4.8: Historical note: the irrationality shock

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: a Pythagorean-style counting pebble arrangement failing to match the diagonal of a square carved into
dark stone.
Conceptual action: The diagonal broke the belief that all magnitudes are ratios of whole numbers; a hairline
crack runs through the neat pebble grid, symbolizing a broken assumption
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: The diagonal broke the belief that all magnitudes are ratios of whole numbers.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_9-A - 4.9: Another lens: ratio as a fingerprint

Placement	macro conceptual image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
```

```
studio light
Subject: two different objects pressed into soft wax, producing the same ratio fingerprint pattern: ridges made
of comparison outcomes rather than literal finger lines.
Conceptual action: A ratio is the invariant comparison pattern shared by three pairs; the physical objects
differ, but the pattern matches perfectly
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: A ratio is the invariant comparison pattern shared by many pairs.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_10-A - 4.10: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a tray of small comparison cards: each card shows stacks of copies being compared, with green/red
tipping results The cards for two ratios line up like a memory game.
Conceptual action: Equal ratio means every comparison test gives matching behavior; reader sees the invitation to
testing more and more cards
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Equal ratio means no test can distinguish them.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-4_11-A - 4.11: What this section teaches beyond geometry

Placement	motif bridge
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a triptych linking Eudoxus comparisons, nested intervals, and calculus limits.
Conceptual action: Mathematics often defines an object by all tests it survives; in each panel, finite
approximations surround a central object that is never touched directly but becomes forced into clarity
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Mathematics often defines an object by all tests it survives.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5-A - 5: Multiplication as scaling, not repeated chanting

Placement	chapter opener

Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a small geometric figure projected by a lantern onto a larger wall, preserving shape while changing
size.
Conceptual action: Multiplication is fundamentally resizing; a second dim shadow made of repeated blocks sits off
to the side as only one limited interpretation
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Multiplication is fundamentally resizing.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_1-A - 5.1: Repeated addition is only one shadow of multiplication

Placement	conceptual split image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: the full object called multiplication as a 3D dilation sculpture casting a flat shadow made of repeated
equal blocks.
Conceptual action: Repeated addition is a special case, one narrow shadow of the meaning; the shadow is useful
but incomplete
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Repeated addition is a special case, not the whole meaning.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_2-A - 5.2: Dilation: the clean geometric meaning

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: points radiating from a fixed center through a pantograph-like mechanism.
Conceptual action: Dilation scales distances from a fixed center; each point moves along its ray farther from the
center by the same scale factor
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Dilation scales distances from a fixed center.

Overlay notes: Overlay optional for center, rays, and scale factor. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_3-A - 5.3: The similar-triangle multiplication machine

Placement	primary technical plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: brass drafting apparatus on translucent vellum
Subject: two rays from one origin, one movable point on the first ray, one height point on the second ray, one parallel guide arm
Conceptual action: show multiplication as a scale transferred by similar triangles
Composition: apparatus only; exact rays, points, and parallel lines will be drawn as vector overlay; leave wide clean margins
Materials / medium: brass pins, taut thread, clear guide arm, vellum layers, soft overhead light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Multiplication is earned by similarity and parallel transfer.

Overlay notes: Vector overlay required for labels, product, and parallel marks. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_3-C - 5.3-C: Similar-triangle multiplication machine - clean overlay plate

Placement	technical-art plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: blank technical construction plate prepared for vector overlay
Subject: empty vellum, origin pin, two ray guides, one movable guide arm, one taut diagonal thread
Conceptual action: provide the physical scaffold for the similar-triangle multiplication machine
Composition: top-down, 70 percent clean vellum, apparatus endpoints clearly visible
Materials / medium: pale vellum, brass pins, graphite smudges, taut linen thread
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The art plate provides atmosphere; the overlay supplies exact labels.

Overlay notes: Add origin, unit, factor, input height, output height, and parallel marks in layout. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_3-B-A - 5.3-B: The similar-triangle multiplication machine - learner close-up

Placement	optional secondary plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled studio light
Subject: the same multiplication idea as a satisfying physical scale copier: a small triangular frame and a larger similar triangular frame connected by synchronized parallel arms.
Conceptual action: The product is the height forced by a similarity machine; a vertical output gauge rises because shape is preserved under scale; exact construction lines are reserved for vector overlay.
Composition: minimal apparatus centered on blank vellum; 70 percent clean overlay area; one construction scaffold only
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension, glass reflections
```

```
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The product is the height forced by a similarity machine.

Overlay notes: Overlay optional for one concrete example. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_4-A - 5.4: Three meanings that textbooks collapse into one symbol

Placement	triptych
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: three glass cases: a scale factor stretching a rope, two lengths forming a rectangle of area, and two
pure scale factors composing like gears.
Conceptual action: The same symbol can package different typed actions; each case uses the same multiplication
glyph as a small removable tag outside the glass, implying one symbol covers different type signatures
Composition: exactly three equal panels with blank gutters; one conceptual role per panel; each panel contains at
most two main objects
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: The same symbol can package different typed actions.

Overlay notes: Overlay for type signatures recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_5-A - 5.5: Why multiplication is commutative

Placement	mirror composition
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: two rays crossing at an origin with two similar-triangle constructions mirrored across a diagonal.
Conceptual action: Scaling by a then b gives the same pure scale as b then a in this setting; the outputs meet at
the same product height from opposite interpretive routes
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Scaling by a then b gives the same pure scale as b then a in this setting.

Overlay notes: Overlay for swapped roles recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_6-A - 5.6: Distributivity becomes geometric

Placement	animated still plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a two-colored ribbon made of segments b and c being enlarged by a dilation.
Conceptual action: Dilation carries joined pieces to joined scaled pieces; after scaling, the join between colors
remains exactly between the scaled pieces
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Dilation carries joined pieces to joined scaled pieces.

Overlay notes: Overlay optional for segment labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_7-A - 5.7: Historical note: geometry and algebra eventually shake hands

Placement	split historical tableau
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: on one side, show Euclidean diagrams on parchment; on the other, algebraic symbols emerging on a
merchant's slate.
Conceptual action: Algebra and geometry became mutually translating languages; in the center, similar triangles
and equations clasp through a mechanical bridge
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Algebra and geometry became mutually translating languages.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-5_8-A - 5.8: Thought experiments

Placement	margin image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a simple shape under three zoom settings: same shape, different sizes, with one marked join inside the
shape remaining proportionally located.
Conceptual action: Imagine resizing the whole world and ask what survives; reader sees the invitation to dragging
the zoom slider and watching every distance scale together
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Imagine resizing the whole world and ask what survives.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6-A - 6: Division as inverse scaling

Placement	chapter opener
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a quiet three-door threshold for division, but distinguish the doors only by action: copies fitting into
a whole, a whole being split into equal shares, and a slope line connecting unlike axes.
Conceptual action: Division is one inverse relation with three different roles; the same small unknown-scale
token sits before all three doors
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Division is one inverse relation with three different roles.

Overlay notes: Overlay optional for type signatures. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_1-A - 6.1: The single equation behind all division

Placement	mechanical metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a machine with input b and output a, but the central scale gear is missing.
Conceptual action: Division asks for the missing transformation; a hand is selecting the gear that would make the
transformation work
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Division asks for the missing transformation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_1-B - 6.1-B: Division roles - restrained triptych

Placement	triptych
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: single tabletop divided into exactly three equal panels
Subject: left: copy-fitting; center: fair splitting; right: slope/rate between unlike axes
Conceptual action: show the three roles of division separated into one panel each
Composition: three panels; one role per panel; each panel has one focal object
Materials / medium: wood table, brass rods, ribbon strip, slope ramp, blank vellum axes
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Division is one inverse relation with three different roles.

Overlay notes: Overlay panel labels later: measurement, sharing, rate. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_2-A - 6.2: Quotitive division: measurement

Placement	horizontal tactile image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a long plank being measured by repeated placement of a smaller standard block.
Conceptual action: Length divided by length produces a pure ratio; the output is a pure count/ratio object; it is
the count/ratio implied by how many times the block fits
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Length divided by length produces a pure ratio.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_3-A - 6.3: Partitive division: fair splitting

Placement	top-down plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a single cloth strip being divided into equal shares by parallel guide lines, with one share lifted
slightly as the answer.
Conceptual action: Length divided by count produces a smaller length; the divisor is implied as a count of
shares, a count of shares rather than a second length
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Length divided by count produces a smaller length.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_4-A - 6.4: Rate division: the missing third type

Placement	cinematic axis metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a train moving across a landscape where the horizontal axis is time and the vertical climb is distance.
Conceptual action: Unlike quantities divide into a new kind of quantity: rate; a glowing diagonal path connects
the origin to the journey point
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
```

```
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension, glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Unlike quantities divide into a new kind of quantity: rate.

Overlay notes: Overlay recommended for axes and units. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_5-A - 6.5: Slope is the geometric picture of rate

Placement	technical plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: two-axis rate table with unlike physical rails
Subject: one horizontal time rail and one vertical distance rail meet at an origin pin; one ramp connects origin to a journey point
Conceptual action: show rate as steepness produced by unlike magnitudes
Composition: one slope ramp only; axes are blank rails, blank physical rails; leave endpoint label area empty
Materials / medium: dark table, brass rails, glass ramp, amber endpoint bead
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Rate is steepness in a plane whose axes carry different units.

Overlay notes: Overlay for sample units recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_5-B-A - 6.5-B: Slope is the geometric picture of rate - alternate

Placement	optional conceptual image
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled studio light
Subject: a mountain trail where every horizontal stride corresponds to time and every vertical rise corresponds to distance.
Conceptual action: The quotient lives as the relation between axes; the same path has a constant incline; the incline itself is highlighted as a new object; exact construction lines are reserved for vector overlay.
Composition: minimal apparatus centered on blank vellum; 70 percent clean overlay area; one construction scaffold only
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension, glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The quotient is not on either axis; it is the relation between axes.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_6-A - 6.6: Why the three cases matter

Placement	triptych corrective
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: the same central division symbol as a neutral mask worn by three different bodies: same-kind
measurement, count-based sharing, unlike-kind rate.
Conceptual action: Type signatures decide what kind of answer division produces; each body has different output
material: pure light token, smaller segment, angled slope ribbon
Composition: exactly three equal panels with blank gutters; one conceptual role per panel; each panel contains at
most two main objects
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Type signatures decide what kind of answer division produces.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_7-A - 6.7: Historical note: where rate changed the game

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: an inclined plane, water clock, rolling ball, and a notebook of measured distances arranged under
moonlit workshop light.
Conceptual action: Physics begins when unlike magnitudes are compared by rate; the visual story is that motion
forced mathematics to divide distance by time, birthing rate as a serious object
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Physics begins when unlike magnitudes are compared by rate.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-6_8-A - 6.8: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: cards showing length/length, length/count, distance/time, mass/volume, and force/area into three trays.
Conceptual action: Ask what kinds of things are being divided before calculating; each tray produces a different
output token
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Ask what kinds of things are being divided before calculating.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7-A - 7: Constructing ratios geometrically

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a handcrafted ratio machine of pins, threads, and parallel arms producing an output height from two
input lengths.
Conceptual action: Division can be constructed by similarity before it becomes a slogan; it feels like a secret
calculator built from pure geometry
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Division can be constructed by similarity before it becomes a slogan.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_1-A - 7.1: The ratio machine

Placement	technical plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: ratio construction apparatus on blank vellum
Subject: two rays, one connector thread, one parallel guide passing through a unit marker position
Conceptual action: show division as finding the scale factor that transfers one magnitude to another
Composition: top-down technical plate; exact geometry added later; apparatus endpoints obvious and uncluttered
Materials / medium: vellum, brass pins, linen thread, clear acrylic parallel guide
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: A quotient is the scale forced by a parallel construction.

Overlay notes: Vector overlay required for a, b, 1, and q labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_2-A - 7.2: What reciprocal actually means

Placement	conceptual metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a camera lens zoomed in by a factor, then a second lens setting that exactly undoes the zoom and returns
the image to original size.
Conceptual action: A reciprocal is an inverse scaling transformation; the reciprocal is visualized as the undo-
setting on the same mechanism, as the undo-setting on the same mechanism
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: A reciprocal is an inverse scaling transformation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_3-A - 7.3: Why dividing by a fraction becomes multiplying by its reciprocal

Placement	process plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: two constructed fractional lengths being fed into the ratio machine, then a separate undo-scaling gear
transforming the divisor back to the unit before scaling the numerator.
Conceptual action: Invert-and-multiply is a theorem about undoing scale, a theorem about undoing scale; it a
clean two-stage causal story: build the lengths, find the inverse scaling
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Invert-and-multiply is a theorem about undoing scale, not a chant.

Overlay notes: Overlay optional for one concrete numerical example. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_3-B-A - 7.3-B: Dividing fractions - alternate anti-slogan

Placement	optional split panel
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: an anti-slogan image: on the left, a hand holds a blank fraction-shaped card about to be flipped, but
the card is shadowy and secondary; on the right, a real inverse-scaling machine rescales one constructed segment
into another using parallel guides.
Conceptual action: The rule is a shadow of inverse scaling; the point is that 'invert and multiply' is only a
shadow of undoing a scaling; exact construction lines are reserved for vector overlay.
Composition: minimal apparatus centered on blank vellum; 70 percent clean overlay area; one construction scaffold
only
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The rule is a shadow of inverse scaling.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_4-A - 7.4: Cross-multiplication is a late shadow, not the meaning

Placement	corrective image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: the rich ratio machine casting a flat cross-shaped shadow on a wall.
```

```
Conceptual action: Cross-multiplication is compressed evidence of a deeper proportion relation; the shadow is
useful and recognizable but visibly downstream from the 3D construction
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: Cross-multiplication is compressed evidence of a deeper proportion relation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_5-A - 7.5: Another lens: zoom factors on a screen

Placement	modern analogy
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: an image on a monitor being resized by a zoom factor.
Conceptual action: A ratio is the zoom factor that makes one magnitude match another; a second image must be
scaled to match the first, and the unknown zoom slider glows
Composition: spacious construction stage; exact theorem lines reserved for vector overlay; one apparatus in focus
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: diagram-art hybrid for geometry-first instruction
```

Caption: A ratio is the zoom factor that makes one magnitude match another.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-7_6-A - 7.6: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: drafting table with translucent vellum, brass pins, taut thread, straightedge, compass, and controlled
studio light
Subject: a map scale bar, a microscope magnification ring, and a blueprint scale model as three quiet objects
sharing the same hidden question: what factor converts one size to another?
Conceptual action: Look for the conversion factor, before the quotient symbol; the shared factor is indicated by
a common amber glow
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: photographed construction apparatus; vellum layers, brass, graphite shadows, thread tension,
glass reflections
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Look for the conversion factor, not just the quotient symbol.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Transformation theatre

GFV6-8-A - 8: Negatives and multiplication

Placement	chapter opener

Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a line through an origin mounted like a kinetic sculpture.
Conceptual action: Negative multiplication is symmetry composition; a half-turn rotation sends a right-pointing
segment to its left-pointing twin; a second half-turn returns it
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Negative multiplication is symmetry composition.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-8_1-A - 8.1: Multiplying by -1 is a geometric action

Placement	subsection plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: multiplying by minus one as a half-turn machine on a line: an arrow enters, passes through a simple
rotating hinge centered at the origin, and exits with the same length in the opposite direction.
Conceptual action: Multiplying by -1 is a half-turn; one ghosted half-turn arc is enough
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Multiplying by -1 is a half-turn.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-8_2-A - 8.2: What (-a)b means

Placement	process image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a segment passing through two stations: a scaling frame that changes its length, then a flipping frame
that reverses orientation.
Conceptual action: Negative scaling combines size change with orientation reversal; a parallel ghost version
flips first then scales, landing at the same final segment
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Negative scaling combines size change with orientation reversal.

Overlay notes: Overlay optional for order notes. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-8_3-A - 8.3: Squares forget orientation

Placement	conceptual metaphor
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: two opposite arrows entering a square-making device and producing identical unoriented square tiles.
Conceptual action: Squaring preserves magnitude but loses sign; the arrows remember left versus right; the tiles
preserve only magnitude
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Squaring preserves magnitude but loses sign.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-8_4-A - 8.4: Historical note: the uneasy status of negatives

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: debt tokens, merchant ledgers, and a number line projected faintly over them.
Conceptual action: Debts made negatives useful; orientation made them coherent; the old objects suggest suspicion
around 'less than nothing'; the geometric projection resolves it as orientation
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Debts made negatives useful; orientation made them coherent.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-8_5-A - 8.5: Thought experiments

Placement	margin visual
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a card with an arrow printed on it being flipped around a central pin once, then twice.
Conceptual action: Count half-turns, half-turn composition rather than sign mnemonics; after two flips it returns
to its original orientation
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Count half-turns, not sign mnemonics.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9-A - 9: Squares and square roots

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a unit square garden expanding equally in two perpendicular directions while a separate number-line
segment scales twice in sequence.
Conceptual action: Squaring means repeated scaling or two-direction area growth depending on type; the two
meanings is visually parallel but kept clearly separated: repeated scaling on a line, area scaling in a plane
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Squaring means repeated scaling or two-direction area growth depending on type.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9_1-A - 9.1: Two legitimate meanings of square

Placement	diptych
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: two panels.
Conceptual action: A square can be a repeated scale factor or an area; left: a segment passes through two
identical scale gates on a line
Composition: exactly two equal panels; left and right are visually parallel; blank gutter between panels; one
action per panel
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A square can be a repeated scale factor or an area.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9_2-A - 9.2: Why squares appear everywhere

Placement	conceptual field image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a city block grid growing outward in two independent directions from a unit square.
Conceptual action: Squares are the fingerprint of two independent directions; every new block belongs to area
created by horizontal and vertical scaling together
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Squares are the fingerprint of two independent directions.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9_3-A - 9.3: Square root as an inverse question

Placement	tactile reconstruction
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: an artisan given a glowing square area rug and using dividers to recover the side length that would
generate it.
Conceptual action: A square root asks for the length that produces an area; the side length is lifted from the
area like an answer extracted from a surface
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A square root asks for the length that produces an area.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9_4-A - 9.4: Why negatives vanish under squaring

Placement	mirror concept
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: two arrows in opposite directions dipped into the same square pool of light; after two flips or two
scalings their orientation stains disappear and only equal magnitude remains.
Conceptual action: Squaring collapses opposite orientations into the same magnitude; be poetic but clear:
direction has been washed out
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Squaring collapses opposite orientations into the same magnitude.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9_5-A - 9.5: Historical note: the diagonal of the square

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: a unit square carved into black stone with its diagonal as a bright metal inlay extending beyond a grid
of rational measuring ticks that fail to catch it.
Conceptual action: The square's diagonal demanded numbers beyond fractions; the diagonal looks undeniable, almost
more real than the ticks
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: The square's diagonal demanded numbers beyond fractions.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-9_6-A - 9.6: Thought experiments

Placement	margin image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: three square tiles of different areas with movable side rods beside them.
Conceptual action: Ask the inverse area question with your hands; one rod must be selected to generate each
square
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Ask the inverse area question with your hands.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Scale-gate and odometer

GFV6-10-A - 10: Exponents as iterated scaling, logarithms as counted resizings

Placement	chapter opener
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: a chain of identical scale gates receding into space.
Conceptual action: Exponents perform repeated resizing; logarithms count the resizings; an object changes size by
the same ratio at each gate
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Exponents perform repeated resizing; logarithms count the resizings.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_1-A - 10.1: The tower of operations

Placement	vertical structural diagram
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick; SYS.4 scale_gate_odometer

```
Scene: one horizontal workbench divided into exactly three stations
Subject: station one shows equal translation steps; station two shows equal resize gates; station three shows an
odometer attached to a resize gate
Conceptual action: show iteration moving from repeated motion to repeated scaling to counting repeated scaling
Composition: three stations, left-to-right causation, blank gutters, horizontal layout
Materials / medium: brass rail, matte blocks, scale gates, blank odometer wheels, soft studio light
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
```

```
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: Each operation level iterates the one below.

Overlay notes: Overlay optional for operation names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_2-A - 10.2: Powers are repeated resizings

Placement	horizontal-to-spiral sequence
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: one object passing through the same resize portal repeatedly.
Conceptual action: A power is one scaling applied again and again; each passage leaves a ghost copy that is
larger by the same ratio than the previous copy
Composition: left-to-right causation; exactly one resizing action or one counting action per image; large
negative space above
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: A power is one scaling applied again and again.

Overlay notes: Overlay can add powers as labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_3-A - 10.3: The exponent laws are structural facts

Placement	mechanical choreography
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean resizing corridor with exactly two choreographies
Subject: left choreography joins two scale-gate chains end-to-end; right choreography repeats one boxed chain as
a unit
Conceptual action: show exponent laws as arrangements of repeated resizing machines
Composition: exactly two panels; each panel carries one law-choreography; open space above each choreography for
overlay
Materials / medium: brushed brass gates, matte ceramic object, black tabletop, soft shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: Exponent laws record how repeated scale actions combine.

Overlay notes: Overlay optional for law names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_3-B - 10.3-B: Exponent law companion - joined chains

Placement	technical-art plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
```

```
crossings
Subject: two chains of identical resize gates joined end-to-end into one longer chain.
Conceptual action: Composing repeated resizings adds the counts; nothing else
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Composing repeated resizings adds the counts.

Overlay notes: Overlay the symbolic law after generation if needed. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_3-C - 10.3-C: Exponent law companion - nested chains

Placement	technical-art plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens; SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: one package containing a short chain of resize gates, then three identical packages repeated in
sequence.
Conceptual action: Repeating an already-repeated resizing multiplies the counts; learner sees see a group of
actions being repeated as a group
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Repeating an already-repeated resizing multiplies the counts.

Overlay notes: Overlay the symbolic law after generation if needed. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_3-D - 10.3-D: Exponent law companion - paired scaling channels

Placement	technical-art plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: an object passing through paired scale channels: each stage applies one horizontal resize and one
vertical resize together, repeated three times.
Conceptual action: Repeated paired resizings can be regrouped when the actions commute; then show the same total
effect as all horizontal resizes gathered and all vertical resizes gathered
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Repeated paired resizings can be regrouped when the actions commute.

Overlay notes: Overlay the symbolic law after generation if needed. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_4-A - 10.4: Fractional exponents are forced, not invented for mischief

Placement	morph sequence
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick; SYS.4 scale_gate_odometer

```
Scene: resize corridor between a starting object and a final target object
Subject: exactly three identical small scale gates fill the gap between start and target
Conceptual action: the highlighted gate is the repeated intermediate resizing that makes the final target
possible
Composition: straight left-to-right path; start, three gates, target; only these five elements
Materials / medium: brass gate frames, matte objects, pale floor track, soft product lighting
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: A fractional power is the scale step that must repeat to make the target.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_4-B - 10.4-B: Fractional exponent companion - the required middle gate

Placement	conceptual plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: a starting object and a final target object with three identical empty gates between them.
Conceptual action: A root is the scale step that must repeat to make the whole scaling; one gate is highlighted
as the required middle resizing: pass through it three times and the target is reached
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: A root is the scale step that must repeat to make the whole scaling.

Overlay notes: Overlay can identify the number of gates later. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_5-A - 10.5: Logarithms count multiplicative steps

Placement	conceptual detective image
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer

```
Scene: close mechanical resize corridor with a small odometer mounted beside the gate path
Subject: one object crosses identical ratio gates while a blank-notch counter advances once per gate
Conceptual action: make logarithm feel like counting how many equal scale steps happened
Composition: counter is the focal object; three scale gates recede behind it; clear blank space for overlay
Materials / medium: brass counter wheels with blank notches, ceramic object, glass gate edges, warm studio
reflections
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: A logarithm asks how many scale steps reach the target.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_5-C - 10.5-C: Logarithm companion - resizing odometer close-up

Placement	margin image
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: a close-up of a beautiful mechanical odometer attached to a resize gate.
Conceptual action: A logarithm counts how many equal ratio steps occurred; each time the gate is crossed, one
small wheel clicks forward
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: A logarithm counts how many equal ratio steps occurred.

Overlay notes: Do not generate digits; overlay any count later. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_5-B-A - 10.5-B: Logarithms count multiplicative steps - alternate

Placement	optional plate
Priority	optional / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick; SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: a minimal cosmic version of the same idea: a small probe crosses concentric ratio shells, each shell
representing one equal multiplicative jump.
Conceptual action: Logarithms measure distance in the resizing world; the four shells are spacious; the probe's
counter glows softly
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Logarithms measure distance in the resizing world.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_6-A - 10.6: The logarithmic line

Placement	technical-art plate
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: a special blank ruler whose spacing represents equal ratios rather than equal differences.
Conceptual action: A logarithmic scale turns multiplicative structure into additive spacing; above it, a curved
growth path is projected down and becomes a straight path on the ratio-spaced ruler
Composition: left-to-right causation; exactly one resizing action or one counting action per image; large
negative space above
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
```

```
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: A logarithmic scale turns multiplicative structure into additive spacing.

Overlay notes: Overlay recommended for a few clean tick labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_7-A - 10.7: Historical note: Napier, tables, and the war against drudgery

Placement	historical note plate
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer; SYS.5 rotating_bead

```
Scene: Napier-era instrument still life; closed books and tools only
Subject: Napier rods, astrolabe, dividers, navigation calipers, compact lookup instrument, and one closed codex
Conceptual action: show logarithms as practical compression of repeated arithmetic labor
Composition: single tabletop still life; compact lookup instrument in focus; tools arranged in a calm arc
Materials / medium: aged brass, dark wood, leather-bound closed book, candlelight, blank instrument faces with
abstract ticks
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Logarithms were invented to turn brutal multiplication into manageable addition.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_7-B - 10.7-B: Napier compression - instruments-only version

Placement	historical companion plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.4 scale_gate_odometer

```
Scene: instrument-only still life on dark wood
Subject: Napier rods, dividers, astrolabe, blank odometer-like lookup cylinder, and one closed leather codex
Conceptual action: show calculation labor compressed into a compact tool
Composition: lookup cylinder centered; rods and instruments form a quiet semicircle
Materials / medium: aged brass, wood, leather, blank ceramic surfaces, candle glow
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Logs began as a machine for removing arithmetic drudgery.

Overlay notes: Use if manuscript/paper surfaces keep producing fake script. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_8-A - 10.8: Why this chapter matters for calculus

Placement	bridge image
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: repeated scale jumps becoming smaller and more frequent until they blur into a smooth growth curve.
Conceptual action: Continuous iterated scaling leads naturally toward calculus; on the smooth side, a tiny
tangent line begins to appear under a refinement lens
Composition: left-to-right causation; exactly one resizing action or one counting action per image; large
```

```
negative space above
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: Continuous iterated scaling leads naturally toward calculus.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-10_9-A - 10.9: Thought experiments

Placement	margin prompt
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick; SYS.4 scale_gate_odometer

```
Scene: clean mechanical resizing corridor with identical scale gates and a physical odometer that counts gate
crossings
Subject: two living growth worlds side by side: one plant gains equal height increments marked by equal spacer
blocks, while another plant passes through equal ratio rings and compounds.
Conceptual action: Equal additions and equal ratios create different worlds; the contrast is visible at a glance:
equal additions versus equal multiplications
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: premium product-style 3D render; brushed brass gates, matte ceramic objects, blank odometer
wheels, soft studio shadows
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Equal additions and equal ratios create different worlds.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Refinement optics

GFV6-11-A - 11: Irrationals, completeness, and the point that survives infinite refinement

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a star being isolated by nested rectangular shutters, each smaller than the last, until one exact point
remains.
Conceptual action: A real number is what survives every refinement; in the foreground, a square diagonal points
to a missing place on a rational ruler
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: A real number is what survives every refinement.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_1-A - 11.1: Density is not the same as no gaps

Placement	conceptual city map

Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a dense city grid full of tiny lit houses between any two houses, but with one glowing empty lot where
an address exists.
Conceptual action: Density differs from completeness; the city is crowded yet incomplete
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Dense does not mean complete.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_2-A - 11.2: Geometry demands sqrt2

Placement	geometric object plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a unit square as a real physical tile with a diagonal brass rod removed and placed against a number line
where The rod looks unquestionably real; the ruler's failure is the problem.
Conceptual action: The diagonal exists before arithmetic can name it neatly.
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: The diagonal exists before arithmetic can name it neatly.

Overlay notes: Overlay optional for square and number-line labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_3-A - 11.3: Nested intervals: defining a point by surviving every refinement

Placement	hero plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a sequence of transparent interval windows closing around one luminous point on a line.
Conceptual action: The point is the unique survivor of all nested constraints; each window is nested inside the
last, with widths shrinking toward invisibility
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: The point is the unique survivor of all nested constraints.

Overlay notes: Overlay optional for interval endpoints. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_3-B-A - 11.3-B: Nested intervals - alternate

Placement	optional plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a glowing bead inside five nested glass boxes, each box thinner than the previous.
Conceptual action: Exactness can be approached by tightening finite cages; the boxes are finite approximations;
the bead is the forced exact object
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Exactness can be approached by tightening finite cages.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_4-A - 11.4: A decimal expansion is a report, not the number

Placement	corrective image
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a stable luminous point on a blank line while an overhead mechanical ticker emits a stream of small
abstract marks that never becomes the main object.
Conceptual action: Digits are reports about the point, separate from the point; the marks are reports; the point
is primary
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Digits are reports about the point, not the point itself.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_5-A - 11.5: Connection back to Eudoxus

Placement	motif callback
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: two-panel bridge on a black optical workbench
Subject: left: two finite balance tests for an unseen ratio; right: nested interval shutters around one unseen
point
```

```
Conceptual action: show the same logic: finite tests force an exact object
Composition: exactly two parallel panels; amber survivor line connects the panels; each panel has one apparatus
Materials / medium: brass balances, glass shutters, black velvet, amber focal line
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Proportion and real numbers both define objects by all finite tests they survive.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_5-B - 11.5-B: Eudoxus-to-completeness callback - same test logic

Placement	motif bridge
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a same-logic callback image: a left hand tests ratios with finite balance comparisons; a right hand
narrows a point with finite interval shutters.
Conceptual action: Ratios, real numbers, and limits all define exactness through finite tests that survive
refinement; between them, one amber survivor line connects the hidden ratio to the hidden point
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Ratios, real numbers, and limits all define exactness through finite tests that survive refinement.

Overlay notes: Overlay arrows or captions in layout only. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_6-A - 11.6: Zeno's shadow and why infinite subdivision does not destroy finiteness

Placement	poetic path image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a ribbon path divided into half, quarter, eighth, and smaller segments, each colored lighter, all
folding back to exactly fill one unit frame.
Conceptual action: Shrinking half-quarter-eighth pieces assemble into one finite whole; the infinity of pieces
looks like refinement, refinement inside the same finite frame
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Infinitely many pieces can still assemble into one finite whole.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_7-A - 11.7: Historical note: rigor took time

Placement	library timeline
Priority	core / regeneration normal

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: ancient geometry manuscripts, calculus notes, and 19th-century analysis texts connected by nested
magnifying lenses.
Conceptual action: Completeness became explicit only after a long historical struggle; suggest that intuition
arrived early but logical housing took centuries to build
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Completeness became explicit only after a long historical struggle.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-11_8-A - 11.8: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a sequence of narrowing calipers around a hidden bead on a line.
Conceptual action: Exactness is the limit of disciplined narrowing; each caliper choice rules out more space
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Exactness is the limit of disciplined narrowing.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Turning and projection observatory

GFV6-A-A - A: Angle, circle, and trigonometric thinking

Placement	interlude opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a rotating telescope arm sweeping a circle under the stars.
Conceptual action: Trigonometry is the mathematics of turning and projected shadow; its tip casts shadows onto
horizontal and vertical measuring walls
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Trigonometry is the mathematics of turning and projected shadow.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_1-A - A.1: Why turning deserves its own chapter

Placement	subsection plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a large compass arm rotating around a fixed point, leaving a luminous arc.
Conceptual action: Turning is a core operation; it is a core operation; nearby, translation and dilation tools
rest on the table, making room for a third basic act: turn
Composition: one turning action; one rotating point and its projections; keep axes and labels for vector overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: Turning is not decoration; it is a core operation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_2-A - A.2: Angle is amount of turn, not just amount of corner

Placement	conceptual image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a lighthouse beam sweeping from one direction to another, leaving an arc of motion.
Conceptual action: An angle measures rotation, the amount of rotation; the wedge between beams is secondary; the
action of turning is primary
Composition: one turning action; one rotating point and its projections; keep axes and labels for vector overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: An angle measures rotation, not merely a static corner.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_3-A - A.3: Radians: the honest unit of angle

Placement	technical-art plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a piece of string equal to the circle's radius being wrapped along the circumference to mark one radian.
Conceptual action: A radian is an angle measured by arc length relative to radius; the radius string and arc
string is visibly congruent, with the central angle formed by the endpoints
Composition: one turning action; one rotating point and its projections; keep axes and labels for vector overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
```

```
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: A radian is an angle measured by arc length relative to radius.

Overlay notes: Overlay recommended for radius, arc, and one-radian mark. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_4-A - A.4: Sine, cosine, and tangent are shape fingerprints

Placement	unit circle diagram-art
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a point on a unit circle with a lamp casting its horizontal and vertical shadows onto two axes, while a
tangent line catches a third shadow.
Conceptual action: Trig functions are projections and ratios attached to a turn; the point, shadows, and tangent
forms one clean visual system
Composition: one turning action; one rotating point and its projections; keep axes and labels for vector overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: Trig functions are projections and ratios attached to a turn.

Overlay notes: Overlay for sin/cos/tan labels recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_4-B-A - A.4-B: Sine, cosine, tangent - alternate

Placement	optional tactile plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a bead moving around a circular track.
Conceptual action: Turning creates changing projections; as it moves, two sliders on perpendicular rails follow
its shadows
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Turning creates changing projections.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_5-A - A.5: Another lens: shadows, slopes, and the unit circle

Placement	horizontal analogy plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
```

```
sky instruments
Subject: the same rotating rod under a low sun, casting a shadow whose length changes, while a slope ramp beside
it rises and falls with the tangent idea.
Conceptual action: Sine and cosine are shadows; tangent is a slope relation; keep the unit circle faint in the
background; foreground the physical shadow
Composition: one turning action; one rotating point and its projections; keep axes and labels for vector overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: Sine and cosine are shadows; tangent is a slope relation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_6-A - A.6: Historical note: trigonometry was forced by the sky

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: an astrolabe, star chart, desert horizon, and ship's navigation instruments arranged around a circular
measuring disk.
Conceptual action: Astronomy and navigation made angle measurement indispensable; the sky is the problem-
generator: distant objects forced humans to measure angles before distances
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Astronomy and navigation made angle measurement indispensable.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_7-A - A.7: Why this interlude pays rent later

Placement	motif bridge
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a three-stage visual bridge: rotating point on circle, its shadow tracing a wave, and the same rotation
becoming multiplication in the complex plane.
Conceptual action: Turning returns as waves, complex multiplication, and periodic motion; use the same bead in
all three stages so the learner sees one object moving through later chapters
Composition: one turning action; one rotating point and its projections; keep axes and labels for vector overlay
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: Turning returns as waves, complex multiplication, and periodic motion.

Overlay notes: Overlay optional for chapter callbacks. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-A_8-A - A.8: Thought experiments

Placement	margin prompt

Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a pendulum swinging while its bob's shadow moves back and forth along a straight track.
Conceptual action: Watch a turn through its shadow; the circular motion and linear projection is visible at once,
inviting the reader to imagine what quantity repeats
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Watch a turn through its shadow.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Transformation theatre

GFV6-12-A - 12: Complex numbers

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a narrow number-line railway reaching a cliff where a plane opens beneath it like a luminous map.
Conceptual action: The real line is too small for rotation; the plane is the right stage; a point leaves the rail
and gains both distance from origin and direction angle
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The real line is too small for rotation; the plane is the right stage.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_1-A - 12.1: Why the real line is not the whole story

Placement	conceptual threshold
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a one-dimensional track that can only move left and right, then reveal a circular turn that cannot live
on the track while preserving connected material the idea.
Conceptual action: Rotation requires a larger space; the plane opens as the missing space where rotation becomes
possible
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
```

```
Use: conceptual book illustration for transformations and functions
```

Caption: Some operations require a larger space.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_2-A - 12.2: A complex number as magnitude and angle

Placement	subsection plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a point in the plane reached by a rope from the origin and a compass arc from the positive axis.
Conceptual action: A complex number stores distance and direction; the rope gives magnitude; the arc gives angle
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A complex number stores distance and direction.

Overlay notes: Overlay optional for magnitude and angle. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_3-A - 12.3: Addition and multiplication are different geometries on the same plane

Placement	diptych
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: two panels on the same complex plane.
Conceptual action: Complex addition moves; complex multiplication rotates and scales; left: arrows slide head-to-
tail for addition
Composition: exactly two equal panels; left and right are visually parallel; blank gutter between panels; one
action per panel
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Complex addition moves; complex multiplication rotates and scales.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_4-A - 12.4: Multiplication becomes multiply magnitudes, add angles

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: two rotate-scale transformations performed in sequence by a dancer-like arrow: first it lengthens and
turns, then lengthens and turns again.
```

```
Conceptual action: Composing rotate-scale actions multiplies lengths and adds turns; a final arrow records
multiplied length and accumulated angle
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Composing rotate-scale actions multiplies lengths and adds turns.

Overlay notes: Overlay for angle arcs recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_5-A - 12.5: What i really is

Placement	iconic plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: i as a small mechanical key that turns every arrow exactly 90 degrees while preserving length.
Conceptual action: i is the unit quarter-turn; one arrow before and after the quarter-turn
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: i is the unit quarter-turn.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_6-A - 12.6: Square roots of negatives

Placement	conceptual plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a half-turn transformation split into two equal quarter-turn arcs.
Conceptual action: The square root of -1 is the operation that turns twice into a half-turn; a rigid arrow
reaches the negative direction after two identical turns
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: The square root of -1 is the operation that turns twice into a half-turn.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_7-A - 12.7: Euler's formula as compressed geometry

Placement	hero plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: turntable observatory with one lamp fixed on a circular rail
Subject: lamp rotates at constant radius while exactly two shadow sliders move on perpendicular rails
Conceptual action: show continuous rotation as bookkeeping: one circular motion, two projected components
Composition: circle centered; two projection rails at right angle; clean empty corner for overlay
Materials / medium: brass turntable, glass lamp, charcoal paper, moonlit shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Euler's formula compresses continuous rotation into exponential language.

Overlay notes: Overlay equation later if desired; do not generate text in image. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Turning and projection observatory

GFV6-12_7-C - 12.7-C: Euler formula companion - turntable bookkeeping

Placement	conceptual plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: observatory table under a dark sky; rotating arm, circular rail, projection walls, shadows, pendulum, and
sky instruments
Subject: a polished turntable with a lamp mounted on its edge.
Conceptual action: Euler's formula is continuous rotation bookkeeping; as the table rotates, the lamp remains at
fixed radius while two shadow sliders move on perpendicular rails
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: night-sky editorial realism; brass pivots, glass bead, charcoal paper, soft moonlight, clean
projected shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Euler's formula is continuous rotation bookkeeping.

Overlay notes: Overlay any formula or angle marks later. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Transformation theatre

GFV6-12_7-B-A - 12.7-B: Euler's formula - alternate

Placement	optional emotional plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a conductor's baton drawing a circle in the air while two soft wave shadows emerge on perpendicular
curtains.
Conceptual action: Rotation projects into sine and cosine; the same motion produces rotation and waves
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Rotation projects into sine and cosine.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_8-A - 12.8: Historical note: from suspicion to indispensability

Placement	historical transformation
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: 'imaginary' roots as ghostly scribbles in an old algebra notebook gradually becoming solid gears inside
electrical circuits, waves, and rotations.
Conceptual action: Complex numbers became real by proving their geometric work; tell the arc from mistrust to
indispensability through instruments and objects
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Complex numbers became real by proving their geometric work.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-12_9-A - 12.9: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a turntable with an arrow painted on it, rotated once by 90 degrees, twice by 180, four times back home.
Conceptual action: Multiply by turning; then count the turns; the setup invites the reader to count turns and
compositions physically
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Multiply by turning; then count the turns.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13-A - 13: Vectors, norms, and matrices

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: arrows rising off the number line into a three-dimensional mobile.
Conceptual action: Directed magnitude grows into transformation geometry; two arrows pass through transformation
frames that stretch, rotate, shear, or flip the surrounding space
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Directed magnitude grows into transformation geometry.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_1-A - 13.1: Vectors are directed segments generalized

Placement	subsection plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a mobile of arrows in 2D and 3D space, all understood as movable directed segments.
Conceptual action: A vector is directed displacement, a movable directed displacement; two arrows are translated
to share a tail with identity preserved
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A vector is directed displacement, not a fixed decorative arrow.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_2-A - 13.2: Magnitude becomes a norm

Placement	technical-poetic plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: three vectors in space with transparent spheres centered at the origin marking equal norm.
Conceptual action: A norm measures length in a space by what rotations preserve; rotating the vectors moves their
components but keeps their tips on the same sphere
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A norm measures length in a space by what rotations preserve.

Overlay notes: Overlay optional for norm notation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_3-A - 13.3: The dot product measures alignment

Placement	diagram-art hybrid
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: one vector as a beam of light projecting another vector's shadow onto its direction.
Conceptual action: The dot product measures how much two directions point together; when vectors align, the
shadow is long; when perpendicular, it nearly disappears
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
```

```
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: The dot product measures how much two directions point together.

Overlay notes: Overlay recommended for projection line. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_4-A - 13.4: Matrices are transformation machines

Placement	hero plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a square mesh entering a transparent transformation machine and exiting as a sheared or rotated mesh.
Conceptual action: A matrix is a machine acting on space; three embedded arrows move with the mesh so the learner
sees that the machine acts on space itself, on space itself before isolated numbers
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A matrix is a machine acting on space.

Overlay notes: Overlay optional for input/output basis vectors. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_4-B-A - 13.4-B: Matrices - alternate

Placement	optional learner plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: the basis vectors as two handles on a flexible grid.
Conceptual action: A linear transformation is determined by where it sends the basis; moving the handles drags
the entire plane with them
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: A linear transformation is determined by where it sends the basis.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_5-A - 13.5: Determinant: area, volume, and orientation change

Placement	technical-art plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a small oriented square of fabric entering a transformation frame and exiting as a parallelogram.
Conceptual action: The determinant measures volume scaling and orientation flip; its area grows or shrinks; if
flipped, its underside color appears
Composition: one machine action; before/after states separated by clear space; exact symbols added later
```

```
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: The determinant measures volume scaling and orientation flip.

Overlay notes: Overlay optional for area factor sign. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_6-A - 13.6: Eigenvectors and eigenvalues: directions the machine treats simply

Placement	conceptual image
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow; SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: seven arrows entering a transformation machine and emerging changed, while one special direction passes
through only stretched or shrunk, still pointing along itself.
Conceptual action: Eigenvectors are directions a transformation keeps its direction; that surviving direction
glows subtly
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Eigenvectors are directions a transformation does not turn.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_6-C - 13.6-C: Eigenvectors companion - direction that survives

Placement	margin image
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow; SYS.5 rotating_bead

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a crowd of arrows entering a shear machine and emerging tilted in five directions, except one arrow that
emerges only longer, still on its original rail.
Conceptual action: An eigenvector is a direction the transformation keeps its direction; the surviving rail glows
amber
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: An eigenvector is a direction the transformation does not turn.

Overlay notes: Overlay name and scaling factor later if desired. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_6-B-A - 13.6-B: Eigenvectors - alternate

Placement	optional plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

Reference tags	SYS.2 amber_invariant_glow

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and controlled spotlights
Subject: arrows in a wind tunnel where most are deflected, but one stream-aligned arrow only lengthens.
Conceptual action: One direction survives a machine as simple scaling; use the wind as the transformation field and the straight survivor as the eigenvector
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Some directions survive a machine as simple scaling.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_7-A - 13.7: Historical note: why this language won

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: engineering diagrams, electrical networks, quantum wave sketches, and computer graphics meshes all feeding into one clean transformation-grid language.
Conceptual action: Linear algebra won because transformation language travels; the point is compression: four practical worlds found the same machinery
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Linear algebra won because transformation language travels.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-13_8-A - 13.8: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and controlled spotlights
Subject: a book rotating around one axis and then another, with a ghost version doing the rotations in the opposite order and ending differently.
Conceptual action: In higher geometry, order can matter; keep the book simple and the axes clear
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: In higher geometry, order can matter.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B-A - B: Coordinates, equations, and translation between shape and symbol

Placement	interlude opener
Priority	core / regeneration normal

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a sculpted curve on one side of a bridge and symbolic machinery on the other.
Conceptual action: Coordinates let geometry and algebra speak to each other; on the bridge, points receive
addresses as coordinate beads
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Coordinates let geometry and algebra speak to each other.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_1-A - B.1: Before coordinates, geometry and algebra spoke different dialects

Placement	historical-conceptual plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a geometer's compass-and-straightedge workspace facing an algebraist's symbolic slate across a narrow
gap.
Conceptual action: Coordinates joined two mathematical languages; a coordinate grid rises between them as a
bridge
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Coordinates joined two mathematical languages.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_2-A - B.2: A point gets an address

Placement	city map analogy
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a lone point in a city grid receiving an address by counting horizontal and vertical moves from a chosen
origin plaza.
Conceptual action: A coordinate is an address relative to a frame; the origin, axes, and unit blocks is visible
as civic infrastructure
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A coordinate is an address relative to a frame.

Overlay notes: Overlay optional for ordered-pair labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_3-A - B.3: Equation as locus, not just a sentence to solve

Placement	hero conceptual plate

Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a cloud of possible points on a dark plane.
Conceptual action: An equation carves a locus out of possible points; an invisible constraint field pulls only
the satisfying points into a glowing curve, like iron filings revealing a magnetic line
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: An equation carves a locus out of possible points.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_4-A - B.4: Solving becomes intersection

Placement	clean visual metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: two glowing paths or rivers crossing in a landscape-grid.
Conceptual action: Solving means finding where constraints agree; the solution is the shared point where both
constraints are true
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Solving means finding where constraints agree.

Overlay notes: Overlay optional for two equations after generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_5-A - B.5: One curve, many descriptions

Placement	triptych
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: the same dancer's ribbon path represented in three adjacent panels: as height over time, as a glowing
set of points satisfying a constraint, and as a timed trail of positions.
Conceptual action: Explicit, implicit, and parametric forms are different access routes to one object; the ribbon
color is identical in all panels to make one curve, three descriptions unmistakable
Composition: exactly three equal panels with blank gutters; one conceptual role per panel; each panel contains at
most two main objects
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Explicit, implicit, and parametric forms are different access routes to one object.

Overlay notes: Overlay optional for description names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_5-B-A - B.5-B: One curve, many descriptions - alternate

Placement	optional plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a single transparent curve-sculpture under three different lights.
Conceptual action: Representations are shadows of one curve; each light projects a different shadow: graph,
contour, and moving trace
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Representations are shadows of one curve.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_6-A - B.6: Another lens: machine view and constraint view

Placement	split conceptual image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: left: a machine accepts inputs and outputs points along a path.
Conceptual action: A curve can be produced by a machine or selected by a constraint; right: a field of possible
points is filtered by a constraint into the same path
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A curve can be produced by a machine or selected by a constraint.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_7-A - B.7: Historical note: analytic geometry changed the landscape

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: a 17th-century desk with maps, axes, geometric curves, and algebraic symbols beginning to overlay one
another.
Conceptual action: Analytic geometry made shape calculable and algebra visible; suggest that placing a grid on
space changed what problems could be seen
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Analytic geometry made shape calculable and algebra visible.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_8-A - B.8: Why this bridge matters for the rest of the book

Placement	road interchange
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: coordinate roads leading forward to function curves, derivative tangents, integrals, and phase
portraits.
Conceptual action: Coordinates are the infrastructure for functions, calculus, and modern geometry; keep the
future topics as distant silhouettes so the current bridge remains focal
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Coordinates are the infrastructure for functions, calculus, and modern geometry.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-B_9-A - B.9: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: one circular ring under three projection lights casting different shadows: vertical graph-like trace,
horizontal footprint, and timed moving dot.
Conceptual action: Change the representation and the same object may become easier to think with; the reader asks
which description is easiest for which question
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Change the representation and the same object may become easier to think with.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14-A - 14: Functions as geometric objects

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a landscape where inputs enter four machines and emerge as curves, surfaces, and transformed objects.
Conceptual action: A function is an object you can operate on; one machine itself is lifted onto a pedestal as
the object of study
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: A function is an object you can operate on.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_1-A - 14.1: The mapping itself is an object

Placement	subsection image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a black-box mapping machine that takes beads from one rail and sends them to another rail according to a
rule.
Conceptual action: The rule itself can become a mathematical object; the entire machine is enclosed in a glass
case as the object being studied
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: The rule itself can become a mathematical object.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_2-A - 14.2: A function does not need a formula

Placement	modern observational plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a weather instrument, a sensor trace, and a set of input-output samples surrounding a sealed machine.
Conceptual action: A function can be known by behavior, by behavior as well as by a closed-form expression; the
mood fights formula worship
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A function can be known by behavior, not just by a closed-form expression.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_3-A - 14.3: The graph makes the function visible

Placement	weaving metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a loom where horizontal input threads and vertical output threads weave into a continuous curve.
Conceptual action: A graph is the visible set of all input-output pairings; each woven knot corresponds to one
input-output pair
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A graph is the visible set of all input-output pairings.

Overlay notes: Overlay optional for axes. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_4-A - 14.4: Composition is feeding one machine into another

Placement	process image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: the output stream of one machine feeding directly into a second machine, with a final output emerging.
Conceptual action: Composition is one mapping fed into another; the connection tube is the visual focus: output
becomes input
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Composition is one mapping fed into another.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_5-A - 14.5: Inverse functions and reflection

Placement	mirror plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a curve approaching a diagonal mirror-door.
Conceptual action: An inverse swaps the direction of the mapping; on the other side, input and output roles have
swapped, producing the inverse curve
Composition: exactly two equal panels; left and right are visually parallel; blank gutter between panels; one
action per panel
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: An inverse swaps the direction of the mapping.

Overlay notes: Overlay optional for y=x mirror line. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_6-A - 14.6: Historical note: Euler and the rise of function-thinking

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: an Euler-era desk where curves rise off paper like living objects and rules, series, and diagrams orbit
them.
Conceptual action: Function-thinking made curves and rules into manipulable objects; the transition is from
solving individual equations to studying families of mappings
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Function-thinking made curves and rules into manipulable objects.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-14_7-A - 14.7: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: four simple machines: one shifts every bead, one stretches spacing, one reverses order, one composes two
actions.
Conceptual action: Ask what the whole machine does, the whole machine before individual outputs; reader sees the
invitation to naming the machine rather than just calculating sample outputs
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Ask what the whole machine does, not just what one input becomes.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15-A - 15: Polynomials

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a curve assembled from modular transformation blocks: shift blocks, scale blocks, power blocks, and
addition rails.
Conceptual action: Polynomials are curves built from operations already earned; the blocks feed into a smooth
polynomial curve, like a sculpture built from known operations
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Polynomials are curves built from operations already earned.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_1-A - 15.1: A polynomial is a constructed transformation

Placement	subsection image
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2
Reference tags	SYS.1 brass_unit_stick

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a polynomial as an assembly line of operations with Each station is distinguished by action, physical
actions.
Conceptual action: A polynomial is an operation recipe made from shifts, scales, and powers.
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
```

```
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A polynomial is an operation recipe made from shifts, scales, and powers.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_2-A - 15.2: Degree tells the far-away story

Placement	landscape curve
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: three polynomial curves as roads leading toward distant horizons.
Conceptual action: The highest-degree term controls the distant shape; higher degree roads dominate the far
distance like mountain ridges, while lower-degree features matter only near town
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: The highest-degree term controls the distant shape.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_3-A - 15.3: Roots are where the machine outputs zero

Placement	mechanical drop gate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a function machine dropping output beads onto a vertical scale; roots are the input positions where
beads land exactly on the zero floor.
Conceptual action: A root is an input that sends the output to zero; the zero floor is a clean horizontal
threshold
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: A root is an input that sends the output to zero.

Overlay notes: Overlay optional for axis and root markers. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_4-A - 15.4: The dimension insight

Placement	border inspection
Priority	core / regeneration medium
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 2

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: polynomial terms at a dimensional border checkpoint.
Conceptual action: Terms can only be added when their dimensions match; length, area, volume, and pure ratio
objects carry visibly different passports made of shape and material, shape and material passports
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
```

```
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Terms can only be added when their dimensions match.

Overlay notes: Overlay optional for type signatures. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_5-A - 15.5: Why polynomials matter so much

Placement	tool chest
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a tool chest containing straightedge, parabola former, cubic spline, and higher-order flexible rulers.
Conceptual action: Polynomials are the simplest flexible tools for approximating smooth behavior; a complicated
curve is being approximated by choosing tools of increasing subtlety
Composition: one machine action; before/after states separated by clear space; exact symbols added later
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual book illustration for transformations and functions
```

Caption: Polynomials are the simplest flexible tools for approximating smooth behavior.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_6-A - 15.6: Historical note: from Babylon to the Renaissance and beyond

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1536x1024 | variants 1

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: babylonian clay tablets, Islamic algebra manuscripts, Renaissance equation notebooks, and modern
graphing traces layered in chronological transparency.
Conceptual action: Polynomial thinking grew across civilizations and notations; a single polynomial curve threads
through all layers
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Polynomial thinking grew across civilizations and notations.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-15_7-A - 15.7: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: small theatrical transformation stage; rails, turntables, levers, mirrors, input/output paths, and
controlled spotlights
Subject: a curve attached to sliders for shifting, stretching, and changing power behavior, but with blank label
zones.
Conceptual action: Tweak the construction and watch the curve respond; moving sliders ghost-changes the curve
shape
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
```

```
for margin placement
Materials / medium: editorial 3D illustration; matte black stage, brass mechanisms, colored glass arrows, soft
smoke-free lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Tweak the construction and watch the curve respond.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Refinement optics

GFV6-C-A - C: Limits, continuity, and convergence

Placement	interlude opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: four blurry paths, sequences, and approximations focusing through a lens onto one sharp object.
Conceptual action: Limits are the discipline of controlled approach; approximation becoming exactness feel like
resolution, a visible guarantee
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Limits are the discipline of controlled approach.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_1-A - C.1: Why approaches needs its own language

Placement	conceptual threshold
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a train slowing toward a station marker through smaller and smaller position gaps.
Conceptual action: Approach needs rules precise enough to trust; the station is the limiting value; the train's
approach is controlled
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Approach needs rules precise enough to trust.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_2-A - C.2: Sequences are itineraries of approximation

Placement	route image
Priority	core / regeneration normal

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: footprints on a path that get closer to a lantern with each step.
Conceptual action: A sequence is a planned list of approximations; each footprint is one approximation; the
lantern determines the itinerary's direction
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: A sequence is a planned list of approximations.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_3-A - C.3: What a limit really promises

Placement	contract image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a target surrounded by rings of tolerance, with all later approximation beads trapped inside any ring
chosen.
Conceptual action: A limit promises eventual containment inside every tolerance; imply a promise: beyond a chosen
threshold, the sequence stays as close as demanded
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: A limit promises eventual containment inside every tolerance.

Overlay notes: Overlay optional for epsilon-style rings. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_4-A - C.4: Continuity: controlled change

Placement	fluid but disciplined
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: fluid moving through a transparent tube whose small input changes cause small output changes.
Conceptual action: Continuity means nearby inputs force nearby outputs; nearby, a cracked jump-tube spills fluid
abruptly
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Continuity means nearby inputs force nearby outputs.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_5-A - C.5: Function limits are about all nearby paths agreeing

Placement	multi-path convergence
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: exactly three paths approaching the same gate from different directions; all must arrive at the same
glowing courtyard.
Conceptual action: A function limit exists only when nearby approaches agree; a single rogue path fading away can
suggest failure while the rest stays sparse
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: A function limit exists only when nearby approaches agree.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_6-A - C.6: Another lens: focus, blur, and guaranteed resolution

Placement	photographic plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a camera viewfinder where a blurry curve sharpens as focus rings turn.
Conceptual action: A limit is guaranteed resolution under refinement; the key is guarantee over aesthetics: every
turn reduces blur according to a controlled mechanism
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: A limit is guaranteed resolution under refinement.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_7-A - C.7: Historical note: from intuition to explicit rigor

Placement	historical note plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: a chalk-dust lecture room where intuitive infinitesimal sketches are being reorganized into precise
nested tolerance bands.
Conceptual action: Calculus intuition eventually required explicit limit language; respect intuition while
showing why rigor was needed
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
```

```
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber focal point, clean shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Calculus intuition eventually required explicit limit language.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_8-A - C.8: Why this interlude pays rent later

Placement	microscope bridge
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine slices
Subject: three refinement scenes connected by one magnifying rail: nested intervals finding a point, secants becoming a tangent, rectangles becoming area.
Conceptual action: Limits are the common grammar behind real numbers and calculus; use the same amber survivor glow in all three
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Limits are the common grammar behind real numbers and calculus.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-C_9-A - C.9: Thought experiments

Placement	margin image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine slices
Subject: beads representing successive approximations eventually staying inside any cage placed around the target.
Conceptual action: Test whether the approximations eventually stay trapped; three cages of different sizes are stacked nearby
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space for margin placement
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Test whether the approximations eventually stay trapped.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16-A - 16: Calculus: the ratio machine at infinite zoom

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine slices
Subject: the earlier ratio machine miniaturized under a microscope on a curve, while below it sixteen thin strips pour into a reservoir of accumulated area.
Conceptual action: Calculus is ratio and accumulation under refinement; a single refinement lens connects derivative and integral
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative space for title overlay
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Calculus is ratio and accumulation under refinement.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_1-A - 16.1: Calculus did not appear from nowhere

Placement	tool genealogy
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine slices
Subject: the tools from earlier chapters - unit rod, parallel lines, ratio machine, log spiral, coordinate grid - all arranged on a wall behind a calculus microscope.
Conceptual action: Calculus is built from earlier acts, assembled from earlier acts; the microscope is assembled from earlier tools and belongs with them
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Calculus is built from earlier acts, not imported from nowhere.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_2-A - 16.2: Derivative: the slope that survives zooming in

Placement	hero plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: microscope over a single curve segment on a glass slide
Subject: one point on the curve and one nearby point approach until the curve looks locally straight
Conceptual action: show derivative as the slope that survives zooming in
Composition: one microscope lens; one local straight segment in focus; surrounding curve fades softly
Materials / medium: glass lens, black slide, amber tangent strip, faint blank vellum grid
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: The derivative is the stable slope revealed by infinite zoom.

Overlay notes: Overlay recommended for secants, tangent, and delta markers. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_2-B-A - 16.2-B: Derivative - alternate

Placement	optional technical plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a tiny similar-triangle ratio machine directly on a curve between two close points.
Conceptual action: The derivative is the ratio machine applied to shrinking changes; as the points move closer,
the apparatus fades into a tangent gauge
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The derivative is the ratio machine applied to shrinking changes.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_3-A - 16.3: Another lens: local linearity

Placement	conceptual landscape
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a rugged coastline seen from far away and through a magnifying glass, where a tiny neighborhood looks
almost straight.
Conceptual action: A smooth curve becomes locally linear when viewed closely enough.
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: A smooth curve becomes locally linear when viewed closely enough.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_4-A - 16.4: The chain rule: rates flowing through nested machines

Placement	process plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: water flowing through one wheel, then another, where each wheel changes the flow rate before passing it
along.
Conceptual action: The chain rule tracks rate through composed machines; the final rate depends on both internal
rates
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
```

```
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: The chain rule tracks rate through composed machines.

Overlay notes: Overlay optional for input/output rate arrows. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_5-A - 16.5: The product rule: two changing sides make one changing area

Placement	technical-art plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a rectangle whose width and height both expand slightly.
Conceptual action: When both sides change, area gains two main strips and a tiny second-order corner; the new
area is decomposed into two side strips plus a tiny corner overlap
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: When both sides change, area gains two main strips and a tiny second-order corner.

Overlay notes: Overlay optional for dx, dy, and area pieces. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_6-A - 16.6: Rates and units

Placement	passport / unit discipline
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: quantities crossing a dimensional border checkpoint: distance/time becomes velocity, area/length becomes
length, velocity*time becomes distance.
Conceptual action: Derivatives divide dimensions; integrals multiply them back in; use passports or tags as
visual devices but
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Derivatives divide dimensions; integrals multiply them back in.

Overlay notes: Overlay for type signatures recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_7-A - 16.7: Integral: accumulation under refinement

Placement	hero plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: accumulation as a landscape under a curve being tiled by vertical slats.
Conceptual action: An integral is accumulated contribution under finer partition; in the foreground the slats are
chunky; farther along they become thinner and thinner until the total shape stabilizes into a single glowing area
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: An integral is accumulated contribution under finer partition.

Overlay notes: Overlay optional for rectangle strips and interval. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_7-B-A - 16.7-B: Integral - alternate

Placement	optional plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: area under a curve as a woven textile made from increasingly thin vertical threads.
Conceptual action: Integration weaves twelve local contributions into one whole; the final cloth is smooth; the
threads remain visible near the edge to show refinement
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Integration weaves many local contributions into one whole.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_8-A - 16.8: The Fundamental Theorem of Calculus

Placement	bridge image
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.3 refinement_lens

```
Scene: two linked workbenches for the same curve
Subject: left bench accumulates thin area slices; right bench extracts the local height from the accumulated area
Conceptual action: show derivative and integral as inverse-ish operations on accumulation and rate
Composition: exactly two benches connected by one loop arrow shape drawn as a physical rail; only these two
benches
Materials / medium: glass slices, matte curve plate, brass rail, warm amber height marker
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: Rate of accumulation returns the thing being accumulated.

Overlay notes: Overlay optional for curve and accumulated area function. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_9-A - 16.9: Taylor series: local information rebuilding the whole

Placement	hero plate

Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.3 refinement_lens

```
Scene: optical mask table focused on one point of a curve
Subject: exactly four transparent masks match value, slope, curvature, and the next bend of the curve
Conceptual action: show Taylor series as local information rebuilding more of the shape
Composition: four masks nested over one focal point; curve visible beneath; open space for overlay
Materials / medium: clear acetate masks, glass pins, amber focal bead, black drafting surface
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: A Taylor series rebuilds a function from local behavior, when convergence allows.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_9-C - 16.9-C: Taylor series companion - local masks

Placement	technical-art plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: a curve near one marked point and four transparent masks stacked nearby: flat value mask, tangent mask,
bending mask, and finer-shape mask.
Conceptual action: Taylor approximation matches more local features, one refinement at a time; each mask hugs the
target curve more closely near the point and loses authority farther away
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Taylor approximation matches more local features, one refinement at a time.

Overlay notes: Overlay degree labels later only if needed. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_9-B-A - 16.9-B: Taylor series - alternate

Placement	optional technical plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: layer transparent curves of degree 0, 1, 2, 3, and higher over a target curve near one point.
Conceptual action: Higher-degree polynomials match more local structure; each layer matches more local features
and extends trust outward
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Higher-degree polynomials match more local structure.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_10-A - 16.10: Historical note: Newton, Leibniz, then rigor

Placement	split historical study
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: newton's motion diagrams and Leibniz's area notation developing on two separate desks, with later
rigorous limit lenses aligning both.
Conceptual action: Calculus emerged from motion, area, and later rigor.
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Calculus emerged from motion, area, and later rigor.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_11-A - 16.11: Common confusions to avoid

Placement	corrective plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: three elegant warning panels: derivative as stabilized ratio, dx as refinement width, integral as
accumulation beyond area contexts.
Conceptual action: The old symbols are safe only when their roles are understood; use visual metaphors -
shrinking secant, unit passport, accumulation reservoir - rather than text-heavy cautions
Composition: one refinement process; exact endpoint or stable survivor centered; surrounding approximations fade
outward
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: pedagogical concept plate for limits, completeness, and calculus
```

Caption: The old symbols are safe only when their roles are understood.

Overlay notes: Overlay warning labels in layout if needed. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-16_12-A - 16.12: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.3 refinement_lens

```
Scene: optical workbench with nested shutters, lenses, microscopes, translucent windows, and increasingly fine
slices
Subject: two small workstations: one with a magnifying lens tightening two curve points into a tangent, the other
with scissors cutting a region into thinner strips.
Conceptual action: Zoom for rate; refine for accumulation; the reader mentally operates both machines
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: macro editorial photography mixed with precise illustration; glass, black velvet, warm amber
focal point, clean shadows
```

```
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Zoom for rate; refine for accumulation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Field and flow

GFV6-D-A - D: Differential equations and waves

Placement	interlude opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-
space traces
Subject: a landscape covered by small arrows indicating local rules of change.
Conceptual action: A differential equation makes the rule of change the object; curves grow by following the
arrows, waves travel across the field, and a phase portrait appears as a hidden map beneath
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank
grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: A differential equation makes the rule of change the object.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_1-A - D.1: A differential equation is a rule about a function and its rates

Placement	subsection image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-
space traces
Subject: a curve traveling through a space where each point carries instructions for its slope or acceleration.
Conceptual action: The equation tells the curve how to change; the curve grows from local rules; it grows by
obeying local rules
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a
time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank
grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: The equation tells the curve how to change.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_2-A - D.2: Slope fields let you see the law before you solve it

Placement	technical-art plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-
space traces
Subject: a field of short line segments like blades of grass tilted by an invisible law.
Conceptual action: A slope field lets the law be seen before formulas solve it; three solution curves weave
through by following the local blade direction
```

```
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: A slope field lets the law be seen before formulas solve it.

Overlay notes: Overlay optional for sample solution curves. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_3-A - D.3: Exponential growth and decay: rate proportional to current amount

Placement	living system metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-space traces
Subject: lanterns whose brightness controls how fast they brighten or fade: bright lanterns change faster, dim lanterns change slower.
Conceptual action: When rate follows current amount, exponential behavior appears; one path grows explosively, another decays gently toward darkness
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: When rate follows current amount, exponential behavior appears.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_4-A - D.4: Harmonic motion: acceleration pointing back toward equilibrium

Placement	motion plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-space traces
Subject: a pendulum or spring displaced from center with acceleration arrows pointing back toward equilibrium, and a ghosted sinusoidal trace below.
Conceptual action: Harmonic motion is change governed by a restoring pull; the motion feels like return, overshoot, return, ordered return, overshoot, return
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: Harmonic motion is change governed by a restoring pull.

Overlay notes: Overlay optional for equilibrium line. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_5-A - D.5: Waves: local coupling creates traveling patterns

Placement	hero plate
Priority	core / regeneration normal

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-space traces
Subject: a line of connected beads or people where one bead pulls the next, creating a wave that travels through the chain.
Conceptual action: A wave is a traveling pattern born from local coupling; the local neighbor interactions clearly, with the global wave emerging across the whole line
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: A wave is a traveling pattern born from local coupling.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_5-B-A - D.5-B: Waves - alternate

Placement	optional emotional plate
Priority	optional / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-space traces
Subject: a stadium wave seen from above dissolving into an ocean ripple diagram.
Conceptual action: The pattern moves; the pieces may only move locally; the same visual rhythm passes through different media, emphasizing that the pattern travels while individual parts mostly oscillate locally
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The pattern moves; the pieces may only move locally.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_6-A - D.6: State space and phase portraits

Placement	conceptual plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-space traces
Subject: instead of graphing height over time, show a dancer's position and velocity as a moving point in a state-space room.
Conceptual action: Sometimes the right picture is the space of states, the state-space path before the time graph; the point traces loops and spirals on the floor
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: Sometimes the right picture is the space of states, not the graph over time.

Overlay notes: Overlay optional for state axes. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_7-A - D.7: Historical note: Newton, Euler, Fourier, and after

Placement	orchestra of laws

Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.5 rotating_bead

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: newton, Euler, Fourier eras as instruments in one orchestra: planets, fluids, strings, heat, and waves
all playing from local rate laws.
Conceptual action: Differential equations became the language of changing physical systems.
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank
grids
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Differential equations became the language of changing physical systems.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_8-A - D.8: Why this interlude matters for the whole mission

Placement	conceptual convergence
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-
space traces
Subject: earlier ideas - rate, function, trig wave, matrix, limit - flowing into a single living landscape
governed by arrows and waves.
Conceptual action: The scaffold becomes powerful when laws of change become objects; the message is that the
book's spine now animates the world
Composition: rule everywhere, path obeying it; exact axes and equations added later; one local law visible at a
time
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank
grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: conceptual science-math plate for differential equations and waves
```

Caption: The scaffold becomes powerful when laws of change become objects.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-D_9-A - D.9: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1

```
Scene: field laboratory; water table, vibrating string, wind tunnel, local arrow markers, currents, and phase-
space traces
Subject: a blank curve starting at one point in an arrow field, with three possible starts elsewhere.
Conceptual action: Follow the local instruction and watch a global curve appear; reader sees the invitation to
drawing a curve by obeying the local arrows, by obeying local arrows before solving equations
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: editorial scientific visualization; translucent fluid, matte arrows, taut string, pale blank
grids
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Follow the local instruction and watch a global curve appear.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Invariant architecture

GFV6-17-A - 17: E = mc^2 and the geometry of invariants

Placement	chapter opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
measuring structures
Subject: two observers slicing the same luminous spacetime crystal at different angles.
Conceptual action: Relativity is invariant geometry in a new symmetry group; their coordinate shadows differ, but
one inner interval glows unchanged
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Relativity is invariant geometry in a new symmetry group.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_1-A - 17.1: Euclidean geometry already trained us for this

Placement	subsection plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
measuring structures
Subject: a vector in the plane viewed through two rotated coordinate grids.
Conceptual action: Rotated observers disagree on components but agree on length; its component shadows change,
but a circular radius ring around the origin preserves the same length
Composition: exactly two observers when observers appear; one invariant object stays fixed; restrained
architectural stillness
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: editorial plate for invariants, relativity, and modern structure
```

Caption: Rotated observers disagree on components but agree on length.

Overlay notes: Overlay optional for x/y components. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_2-A - 17.2: Relativity changes the symmetry group

Placement	hero conceptual plate
Priority	core / regeneration high
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 3
Reference tags	SYS.2 amber_invariant_glow

```
Scene: translucent architectural prism standing on a stone table
Subject: exactly two observers place different slicing frames through the same prism while one amber core line
remains unchanged
Conceptual action: show relativity as different decompositions of one invariant structure
Composition: two observers only; prism centered; amber invariant line uninterrupted through both slices
Materials / medium: glass prism, graphite stone, brass slicing frames, restrained architectural light
```

```
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: editorial plate for invariants, relativity, and modern structure
```

Caption: Relativity changes which transformations preserve structure.

Overlay notes: Overlay for ct and x axes recommended. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_2-C - 17.2-C: Relativity companion - observer slices of one crystal

Placement	hero plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
measuring structures
Subject: one translucent spacetime crystal with two calm observer slices cutting it at different angles.
Conceptual action: Observers disagree about components but preserve the interval; the slices have different
shadows on the outside frame, but one internal amber interval is identical in both views
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: Observers disagree about components but preserve the interval.

Overlay notes: Overlay observer labels and interval line after generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_2-B-A - 17.2-B: Relativity symmetry - alternate

Placement	optional plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
measuring structures
Subject: two measuring frames sliding past the same spacetime event-pair.
Conceptual action: Coordinates flex while the invariant holds fixed; their time and space shadows differ, but
both frames lock onto the same amber interval core
Composition: minimal companion plate; one apparatus centered; at least half the image is clean overlay space
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The coordinates flex; the invariant does not.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_3-A - 17.3: Energy and momentum form a spacetime-like object

Placement	technical-poetic plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
```

```
measuring structures
Subject: energy and momentum as components of a luminous vector inside a dark prism.
Conceptual action: Mass is tied to the invariant length of energy-momentum; as observer frames tilt, the
components slide along different faces, but the vector's invariant core remains the same
Composition: exactly two observers when observers appear; one invariant object stays fixed; restrained
architectural stillness
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: editorial plate for invariants, relativity, and modern structure
```

Caption: Mass is tied to the invariant length of energy-momentum.

Overlay notes: Overlay optional for rest-frame and momentum component labels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_4-A - 17.4: Why this belongs in the same book

Placement	summary plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow

```
Scene: invariant archive room with exactly six drawers arranged in one row
Subject: each drawer contains a simple object transformed by a symmetry; one amber thread runs through all
drawers
Conceptual action: show the same scaffold returning across number line, plane, scaling, complex plane, spacetime,
and energy-momentum
Composition: six drawers, equal size, blank fronts; one drawer slightly open; remaining drawers closed and calm
Materials / medium: matte stone archive, brass handles, amber thread, translucent glass covers
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: editorial plate for invariants, relativity, and modern structure
```

Caption: The same invariant habit runs from segments to spacetime.

Overlay notes: Overlay optional for domain names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_4-B - 17.4-B: Invariant scaffold - one object through six symmetries

Placement	companion recap plate
Priority	optional / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: long stone measurement hall with exactly six identical glass chambers
Subject: one amber measuring thread passes through six transformed versions of a simple object
Conceptual action: show sameness surviving different symmetry chambers
Composition: six chambers in one row; one object per chamber; thread is the only continuous element
Materials / medium: stone, glass, brass frames, amber thread, quiet architectural lighting
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: companion plate designed to receive exact vector diagram overlays
```

Caption: The invariant is what survives a change of viewpoint or transformation.

Overlay notes: Add the six domain labels later as vector type; keep drawer fronts blank in generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_5-A - 17.5: Historical note: Einstein, Minkowski, and the geometric upgrade

Placement	historical transition
Priority	core / regeneration medium

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: instrument-led historical still life; one era-specific work surface, closed books, blank tools only
Subject: einstein and Minkowski historically through objects rather than portraits: train platform, light clock,
geometric spacetime notebook, and a crystalline interval diagram becoming solid.
Conceptual action: Once the right geometry appeared, the formulas became structural; the historical upgrade is
from event timing to geometry
Composition: single still-life or one restrained historical scene; exactly one practical problem is visible
through instruments
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: closed books, blank paper, abstract ticks only; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: historical editorial plate; context comes from instruments and one practical problem; restrained scene
```

Caption: Once the right geometry appeared, the formulas became structural.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_6-A - 17.6: A brief modern echo: Noether

Placement	beautiful theorem metaphor
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
measuring structures
Subject: a symmetry mechanism connected to conserved reservoirs: when a rotation gear turns smoothly, an amber
reservoir remains level; when a translation rail slides smoothly, another reservoir remains level.
Conceptual action: Symmetry and conservation are two sides of one structure; the conserved quantities is shown as
stable physical levels, physical actions
Composition: exactly two observers when observers appear; one invariant object stays fixed; restrained
architectural stillness
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: editorial plate for invariants, relativity, and modern structure
```

Caption: Symmetry and conservation are two sides of one structure.

Overlay notes: Overlay optional for conservation names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-17_7-A - 17.7: Thought experiments

Placement	margin prompt
Priority	core / regeneration normal
API	gpt-image-1.5 | quality medium | size 1024x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow

```
Scene: quiet architectural prism room; observer slices, symmetry frames, amber preserved core, and stone/glass
measuring structures
Subject: two observers with tilted spacetime grids measuring the same pair of events.
Conceptual action: Ask what all observers agree on; their coordinate projections differ, but a central luminous
interval connects the events unchanged
Composition: interactive puzzle-like still life; one object and one visible manipulation; generous empty space
for margin placement
Materials / medium: restrained architectural editorial render; translucent glass, graphite stone, amber thread,
sober studio light
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: small margin plate for reflective adult learning
```

Caption: Ask what all observers agree on.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Modern map and recovery

GFV6-18-A - 18: Where to go next: strengthening the joints

Placement	chapter opener
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: the completed mathematical scaffold with seven joints glowing for future reinforcement: real numbers,
algebraic laws, Euler rotation, differential fields, linear transformations, probability, and topology.
Conceptual action: Further study reinforces the same spine; each future path is a doorway with a distinct visual
action, a set of distinct visual actions
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Further study reinforces the same spine.

Overlay notes: Overlay optional for destination names. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_1-A - 18.1: Build the real numbers rigorously

Placement	subsection image
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.3 refinement_lens

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: two construction crews completing the same bridge over a gorge in the number line: one crew builds with
nested cuts, the other with converging sequences of stepping stones.
Conceptual action: Rigor gives every convergent process a home; they meet at one completed real-line span
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Rigor gives every convergent process a home.

Overlay notes: Overlay optional for two construction methods. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_2-A - 18.2: Re-derive algebraic laws from the geometry

Placement	evidence wall
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: algebraic laws as sealed case files connected by strings to geometric constructions: parallelogram,
dilation, reciprocal, half-turn.
Conceptual action: A rule becomes trustworthy when you can locate its geometry; the point is that every rule has
```

```
evidence, evidence rather than authority
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: A rule becomes trustworthy when you can locate its geometry.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_3-A - 18.3: Go deeper on Euler's formula and complex exponentials

Placement	destination plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a rotating circular lamp whose shadows become waves while a scale-gate mechanism feeds the turn
continuously.
Conceptual action: Euler's formula is where turning, waves, and exponentials meet; this is a promise of a deeper
Euler chapter: addition of angle becomes multiplication of rotations
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Euler's formula is where turning, waves, and exponentials meet.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_4-A - 18.4: Partial differential equations, fields, and nonlinear dynamics

Placement	landscape plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a field landscape where heat flows, fluid swirls, and waves propagate through local arrows.
Conceptual action: Local laws over two spatial variables plus time create rich global behavior; in one corner, a
simple rule produces a complex chaotic braid
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Local laws over many variables create rich global behavior.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_5-A - 18.5: Linear algebra as the grown-up form of transformation geometry

Placement	cockpit metaphor
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: a control room for space transformations: basis levers, eigen-direction monitors, determinant gauges, diagonalization panels.
Conceptual action: Linear algebra is transformation geometry made systematic; the grid outside responds to every control
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Linear algebra is transformation geometry made systematic.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_6-A - 18.6: Probability and statistics as structured uncertainty

Placement	conceptual terrain
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick; SYS.2 amber_invariant_glow

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: uncertainty as fog over a landscape, with measurement tools extracting expectation, spread, correlation, and information channels.
Conceptual action: Probability studies structured uncertainty with the same habits of measure and invariant; the fog means structured uncertainty rather than blank ignorance
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Probability studies structured uncertainty with the same habits of measure and invariant.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-18_7-A - 18.7: Topology and geometry beyond rigid shape

Placement	rubber-sheet art
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: a coffee cup, donut, and looped clay object smoothly deforming on a rubber sheet while holes and connectedness glow as preserved features.
Conceptual action: Change the allowed transformations and 'same' changes too; distances stretch freely; topological invariants remain lit
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Change the allowed transformations and 'same' changes too.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19-A - 19: A modern continuation

Placement	chapter opener

Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: the mathematical cathedral extending beyond the calculus nave into modern wings: algebra, topology,
differential geometry, analysis, probability, computation.
Conceptual action: The scaffold survives into modern mathematics; the same foundation beams from the early
chapters run under all wings
Composition: chapter-opener plate; one dominant metaphor with two or three supporting objects; wide negative
space for title overlay
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The scaffold survives into modern mathematics.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_1-A - 19.1: Abstract algebra: symmetry becomes the object itself

Placement	subsection image
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.1 brass_unit_stick

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: four transformations first acting on a shape, then stepping away from the shape to form their own
choreography circle.
Conceptual action: Abstract algebra studies the structure of operations themselves; the composition law becomes
the dance itself
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Abstract algebra studies the structure of operations themselves.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_2-A - 19.2: Topology: sameness under continuous deformation

Placement	conceptual clay plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a shape being stretched, bent, and smoothed while preserving connected material, while a glowing hole
remains protected through every deformation.
Conceptual action: Topology changes the sameness rule to continuous deformation; correct the notion that topology
is weirdness for its own sake
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Topology changes the sameness rule to continuous deformation.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_3-A - 19.3: Differential geometry: calculus on curved spaces

Placement	hero modern plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: tiny explorers on a curved surface carrying tangent planes, measuring geodesic paths, and detecting
curvature with local straight rulers.
Conceptual action: Differential geometry makes local linear thinking work on curved spaces; the surface feels
like a living extension of calculus onto bent space
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Differential geometry makes local linear thinking work on curved spaces.

Overlay notes: Overlay optional for tangent plane and geodesic. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_4-A - 19.4: Analysis: rigor about limit, approximation, and continuity

Placement	laboratory plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: functions passing through a series of testing chambers: convergence, continuity, completeness,
compactness.
Conceptual action: Analysis formalizes the approximation habits used throughout the book; each chamber uses the
same refinement motif from Interlude C, now sharpened into a lab apparatus
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Analysis formalizes the approximation habits used throughout the book.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_5-A - 19.5: Probability, information, and computation

Placement	modern conceptual triptych
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: three linked panels: probability as structured fog, information as compression into a bright channel,
computation as a path through a maze of allowed procedures.
Conceptual action: Modern mathematics also studies uncertainty, compression, and process; a common thread of
measurement and transformation connects all three
Composition: exactly three equal panels with blank gutters; one conceptual role per panel; each panel contains at
most two main objects
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
```

```
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: Modern mathematics also studies uncertainty, compression, and process.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_6-A - 19.6: Why this modern map matters for a recovering learner

Placement	emotional architecture plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a narrow corridor of rote drills opening into a spacious mathematical structure where comparison,
transformation, refinement, and invariant appear as visible beams.
Conceptual action: The learner was shown the hallway before the cathedral; the beams are distinguished by objects
and actions, physical actions
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: The learner was shown the hallway, not the cathedral.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-19_7-A - 19.7: Final thesis

Placement	closing hero plate
Priority	core / regeneration medium
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: the final thesis as a spiral ascent made of recurring actions: unmeasured segment, chosen unit, directed
path, partition, ratio test, scale gate, turntable, mapping machine, refinement lens, field flow, invariant
crystal.
Conceptual action: The methods change; the spine stays fixed; keep it elegant and legible, with one continuous
amber thread
Composition: organized stations or one central object; scan left-to-right using the exact station count required
by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral
palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: The methods change; the spine does not.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-APP_A-A - APP.A: Historical timeline at a glance

Placement	appendix opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread
tying ideas together
Subject: a horizontal river of mathematical history flowing from ancient measurement through Greek proof,
algebra, analytic geometry, calculus, rigor, relativity, and modern abstraction.
```

```
Conceptual action: History is a sequence of pressures that forced better concepts; each era is a small landscape island with one visual motif from the book
Composition: wide left-to-right route with exactly seven milestones; each milestone is visually simple; open sky/space above for captions
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: History is a sequence of pressures that forced better concepts.

Overlay notes: Overlay dates and names in layout. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-APP_B-A - APP.B: A compact glossary of recurring ideas

Placement	appendix opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: a well-organized cabinet with drawers for magnitude, unit, ratio, transformation, invariant, function, limit, rate, and symmetry.
Conceptual action: The glossary is a cabinet of reusable tools; each drawer contains a small object-motif from the chapters
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: The glossary is a cabinet of reusable tools.

Overlay notes: Overlay drawer labels after generation. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-APP_C-A - APP.C: A small reading discipline for this book

Placement	appendix opener
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: a calm adult study desk with four passes arranged as physical objects: sketch pad, action arrows, relation thread, compressed notation card.
Conceptual action: Read by moving between picture, action, relation, and compression; a bookmark moves through the book in layers, as layered reading rather than a single straight pass
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: Read by moving between picture, action, relation, and compression.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-APP_D-A - APP.D: How to read a proof without feeling ambushed

Placement	appendix opener
Priority	core / regeneration normal

API	gpt-image-1.5 | quality high | size 1536x1024 | variants 2

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: a proof as a route through a landscape: assumptions as starting camp, lemmas as bridges, conclusion as summit.
Conceptual action: A proof is a controlled route from assumptions to conclusion; beside it, evidence tokens are checked into place
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: opening editorial spread for a geometry-first adult learning book
```

Caption: A proof is a controlled route from assumptions to conclusion.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-CLOSE-A - CLOSE: Closing note

Placement	final small plate
Priority	core / regeneration normal
API	gpt-image-1.5 | quality high | size 1536x1024 | variants 1

```
Scene: calm recovery-map studio; modular wall, cabinets, paths, cards, instruments, and a single amber thread tying ideas together
Subject: the reader's tools - unit rod, compass, lens, thread, and notebook - placed at the threshold of an open landscape where curves, waves, and geometric structures continue into the distance.
Conceptual action: The book ends by opening the next field of view; any paper, slate, card, ticker, or manuscript surfaces uses abstract non-readable marks only
Composition: organized stations or one central object; scan left-to-right using the exact station count required by the subject; clear gutters between zones
Materials / medium: museum-quality editorial illustration; paper, wood, brass, glass, graphite, warm neutral palette
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: The book ends by opening the next field of view.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-SYS_1-A - SYS.1: Recurring callback: the brass unit stick

Placement	production reference
Priority	system / regeneration normal
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 3
Reference tags	SYS.1 brass_unit_stick

```
Scene: single-object product reference shot on warm neutral tabletop
Subject: one brass unit stick centered horizontally; beveled ends; two small edge nicks; matte patina; no markings
Conceptual action: create the reusable reference asset for measurement, unit choice, and early ratio scenes
Composition: stick fills the center third of the image; clean empty border on all sides; straight top-down view
Materials / medium: aged brass, subtle scratches, warm wood tabletop, soft side shadow
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: A stable visual prop for unit choice and measurement.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-SYS_2-A - SYS.2: Recurring callback: amber invariant glow

Placement	production reference

Priority	system / regeneration medium
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow

```
Scene: single motif reference on dark matte stone
Subject: one thin amber thread passing through a small glass bead, glowing softly where it crosses the bead
Conceptual action: create the reusable visual marker for “what remains unchanged” across the book
Composition: thread runs left-to-right through center; bead centered; large clean border for cropping
Materials / medium: amber glass light, graphite stone, restrained glow, sharp contact shadow
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: One lighting motif marks what remains unchanged.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

GFV6-SYS_3-A - SYS.3: Recurring callback: refinement lens

Placement	production reference
Priority	system / regeneration medium
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 2
Reference tags	SYS.2 amber_invariant_glow; SYS.3 refinement_lens

```
Scene: single refinement-lens reference asset on black velvet
Subject: one brass-rimmed glass lens hovering above nested rectangular shutters that close toward one point
Conceptual action: create the reusable visual marker for limits, refinement, convergence, and local zoom
Composition: lens centered; exactly five nested shutters; focal point under lens; clean border on all sides
Materials / medium: clear glass, brass rim, black velvet, amber focal point, soft reflected highlights
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: front-matter or recap plate for an adult learning book
```

Caption: One lens motif ties every limit process together.

Overlay notes: Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Scale-gate and odometer

GFV6-SYS_4-A - SYS.4: Recurring callback: scale gate and odometer

Placement	production reference
Priority	system / regeneration medium
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 2
Reference tags	SYS.4 scale_gate_odometer

```
Scene: single-object product reference shot in a clean studio
Subject: one brass resize gate with one blank-notch odometer wheel attached to its right side
Conceptual action: create the reusable reference asset for repeated scaling and logarithm counting
Composition: gate centered, odometer attached, clean border on all sides, straight-on three-quarter view
Materials / medium: brushed brass, matte ceramic base, blank wheel notches, soft studio reflection
Surface / text policy: Text layer: odometer wheels use blank notches or light positions; generated image contains
zero letters, numerals, formulas, signage, or pseudo-script.
Use: editorial instructional plate for exponents and logarithms
```

Caption: One scale gate and counter motif anchors resizing chapters.

Overlay notes: Use as reference input for Section 10 images. Preserve gate proportions and blank counter wheels. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.

Turning and projection observatory

GFV6-SYS_5-A - SYS.5: Recurring callback: rotating bead on circular rail

Placement	production reference
Priority	system / regeneration medium
API	gpt-image-1.5 | quality high | size 1024x1024 | variants 2
Reference tags	SYS.5 rotating_bead

```
Scene: single-object observatory reference shot on charcoal paper
Subject: one glass bead mounted on a brass circular rail with one small pivot at the center
Conceptual action: create the reusable reference asset for rotation, projection, periodicity, and Euler motion
Composition: circular rail centered; bead at upper-right quadrant; large clean border for cropping
Materials / medium: brass rail, glass bead, charcoal paper, soft moonlit shadow
Surface / text policy: Text layer: all surfaces are blank or abstract texture; generated image contains zero
letters, numerals, formulas, signage, or pseudo-script.
Use: book plate for trigonometry, periodicity, and rotation
```

Caption: One rotating bead motif anchors turning chapters.

Overlay notes: Use as reference input for trigonometry and complex-number images. Preserve bead, rail radius, and shadow logic. Add exact labels, axes, theorem lines, tick marks, and equations only as vector overlays after generation.